

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df(data):
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    # Removing of the columns where there is more than 15% of nan
    #df_features = pd.DataFrame(features_all)
    cols_to_check = target_col + features_X
```

```

nan_ratio = df[cols_to_check].isna().mean()
valid_features = nan_ratio[nan_ratio <= 0.15].index
valid_features_list = list(valid_features)

#df = df[valid_features]
# Removing of last np.nan
mask = np.isfinite(df[valid_features]).all(axis=1)
df = df[mask]

cols = list(dict.fromkeys(valid_features_list + #target_col + features_X +
                          feats_to_balance + feats_grouping +
                          feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

new_feats_X = [x for x in features_X if x in valid_features_list]
new_feats_add = [x for x in feats_eng if x in valid_features_list]

data["df"]=df
data["features_X"] = new_feats_X
data["feats_added"] = new_feats_add

column_removed = [x for x in cols_to_check if x not in valid_features_list]
print(f'Column removed because np.isnan > 15%: {column_removed}')

return data

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":

```

```

df["campaign"] = df["datetime"].apply(assign_campaign)
df = df[df["campaign"] != "other"].copy()
df["innova_point"] = df.apply(assign_innova_point, axis=1)
df = df[df["innova_point"] != "no"].copy()
with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r") as f:
    #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r") as f:
        dict_features = json.load(f)
elif target_col[0] == "i_CO2":
    df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
    df = df[df["campaign"] != "other"].copy()
    df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
    df = df[df["innova_point"] != "no"].copy()
    with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
        dict_features = json.load(f)
else:
    raise ValueError("The target col do not match any existing function")

df["hot"] = df.apply(assign_hot_season,axis=1)
df["umidity_range"] = df.apply(assign_humidity, axis = 1)
#Removing of rabbit 4
df["columnn"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features

```

```

def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}

```

```

        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")

```

Column removed because np.isnan > 15%: ['r_temperature_fft_6h_fft_f2_cph', 'r_temperature_fft_6h_fft_period2_h', 'r_temperature_fft_6h_fft_power2', 'r_temperature_fft_6h_fft_f3_cph', 'r_temperature_fft_6h_fft_period3_h', 'r_temperature_fft_6h_fft_power3', 'r_umidity_fft_6h_fft_f3_cph', 'r_umidity_fft_6h_fft_period3_h', 'r_umidity_fft_6h_fft_power3']

1.2 Metadata

Nombre de lignes df	6821
Nombre de lignes df clean	6821
Nombre de features	23
% de NaN	0.0
Taille dataset entraînement	4774
Taille dataset test	2046
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_fft_6h_fft_f1_cph, r_NH3_fft_6h_fft_period1_h, r_NH3_fft_6h_fft_power1, r_NH3_fft_6h_fft_f2_cph, r_NH3_fft_6h_fft_period2_h, r_NH3_fft_6h_fft_power2, r_NH3_fft_6h_fft_f3_cph, r_NH3_fft_6h_fft_period3_h, r_NH3_fft_6h_fft_power3, r_temperature_fft_6h_fft_f1_cph, r_temperature_fft_6h_fft_period1_h, r_temperature_fft_6h_fft_power1, r_umidity_fft_6h_fft_f1_cph, r_umidity_fft_6h_fft_period1_h, r_umidity_fft_6h_fft_power1, r_umidity_fft_6h_fft_f2_cph, r_umidity_fft_6h_fft_period2_h, r_umidity_fft_6h_fft_power2

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

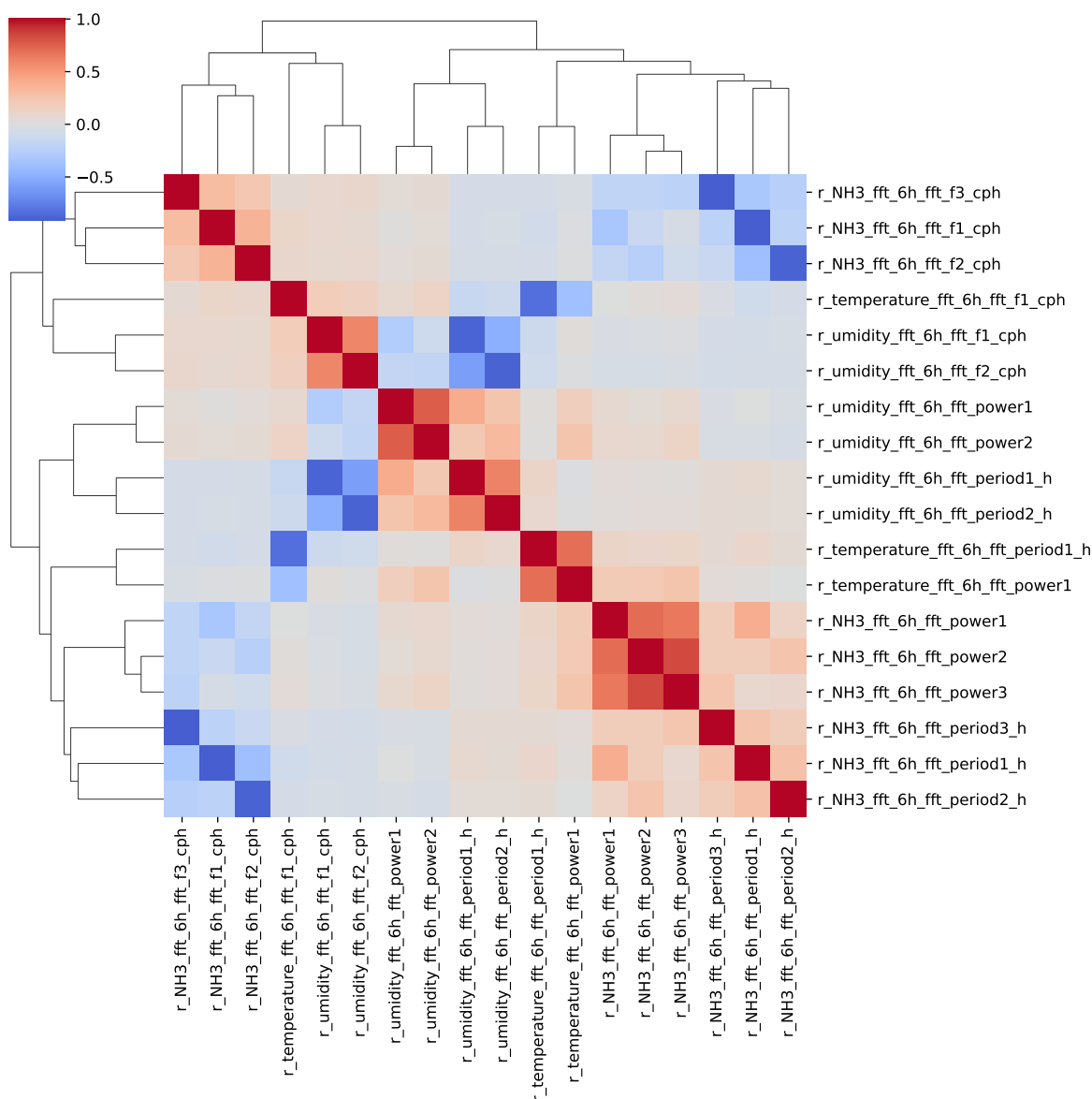
The following section has been done with a threshold choosen of : 0.8

cluster 1: r, , N, H, 3 ; **cluster 2:** r, , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, , u, m, i, d, i, t, y ; **cluster 4:** r, , C, O, 2 ; **cluster 5:** r, , T, H, I ; **cluster 6:** r_NH3_fft_6h_fft_f1_cph, r_NH3_fft_6h_fft_period1_h ; **cluster 7:** r_NH3_fft_6h_fft_power1 ; **cluster 8:** r_NH3_fft_6h_fft_f2_cph, r_NH3_fft_6h_fft_period2_h ; **cluster 9:** r_NH3_fft_6h_fft_power3, r_NH3_fft_6h_fft_power2 ; **cluster 10:** r_NH3_fft_6h_fft_f3_cph, r_NH3_fft_6h_fft_period3_h ; **cluster 11:** r_temperature_fft_6h_fft_period1_h, r_temperature_fft_6h_fft_f1_cph ; **cluster 12:** r_temperature_fft_6h_fft_power1 ; **cluster 13:** r_umidity_fft_6h_fft_f1_cph, r_umidity_fft_6h_fft_period1_h ; **cluster 14:** r_umidity_fft_6h_fft_power1 ; **cluster 15:** r_umidity_fft_6h_fft_period2_h, r_umidity_fft_6h_fft_f2_cph ; **cluster 16:** r_umidity_fft_6h_fft_power2 ;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Choosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_fft_6h_fft_f1_cph, r_NH3_fft_6h_fft_power1, r_NH3_fft_6h_fft_f2_cph, r_NH3_fft_6h_fft_power3, r_NH3_fft_6h_fft_f3_cph, r_temperature_fft_6h_fft_period1_h, r_temperature_fft_6h_fft_power1, r_umidity_fft_6h_fft_f1_cph, r_umidity_fft_6h_fft_power1, r_umidity_fft_6h_fft_period2_h, r_umidity_fft_6h_fft_power2



3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_fft_6h_fft_f1_cph, r_NH3_fft_6h_fft_power1, r_NH3_fft_6h_fft_f2_cph, r_NH3_fft_6h_fft_power3, r_NH3_fft_6h_fft_f3_cph, r_temperature_fft_6h_fft_period1_h, r_temperature_fft_6h_fft_power1,

r_umidity_fft_6h_fft_f1_cph, r_umidity_fft_6h_fft_power1, r_umidity_fft_6h_fft_period2_h, r_umidity_fft_6h_fft_power2

Len(Features_X): 16

Config Gridsearch:{‘show’: True, ‘search_type’: ‘optuna’, ‘model_name’: None, ‘model’: None, ‘param_grid’: None, ‘best_param’: None, ‘cv_func’: ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), ‘save_models’: True, ‘sumup-table’: True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_{\text{test}} < 0.95$ (under performance of the test)

[Green] $r_{\text{test}} > 1.05$ (over performance of the test)

[Red] r_{train} not in $[0.95, 1.05]$ (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_train vs X_test

External Split: Train: 4774 samples Test: 2047 samples



3.3.2 Internal CV Folds

Fold 1/5

Train: 3341 samples Test: 1433 samples Excluded: 0 samples Window exclusion: 0h



Fold 2/5

Train: 3341 samples Test: 1433 samples Excluded: 0 samples Window exclusion: 0h



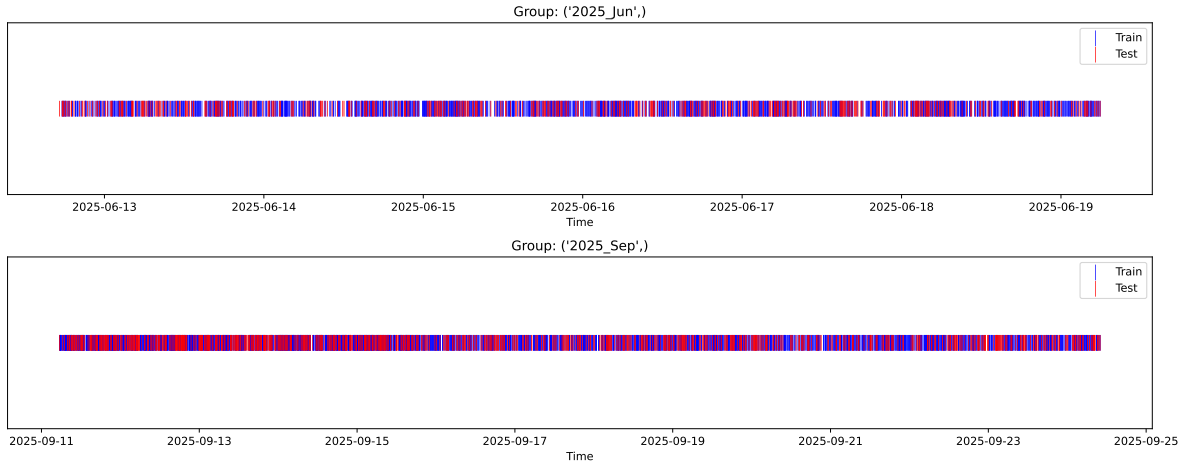
Fold 3/5

Train: 3341 samples Test: 1433 samples Excluded: 0 samples Window exclusion: 0h



Fold 4/5

Train: 3341 samples Test: 1433 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3341 samples Test: 1433 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.315, std=0.761, min=0.973, max=7.779
 y_{test} : mean=2.354, std=0.792, min=0.959, max=8.190

Fold 0 — y_{test} : mean=2.32, std=0.78, y_{train} : mean=2.31, std=0.75

Fold 1 — y_{test} : mean=2.29, std=0.73, y_{train} : mean=2.32, std=0.77

Fold 2 — y_{test} : mean=2.30, std=0.75, y_{train} : mean=2.32, std=0.77

Fold 3 — y_test: mean=2.32, std=0.74, y_train: mean=2.31, std=0.77

Fold 4 — y_test: mean=2.31, std=0.75, y_train: mean=2.32, std=0.76

train scores CV: [0.37437892 0.36180664 0.36057048 0.3564914 0.3659552]

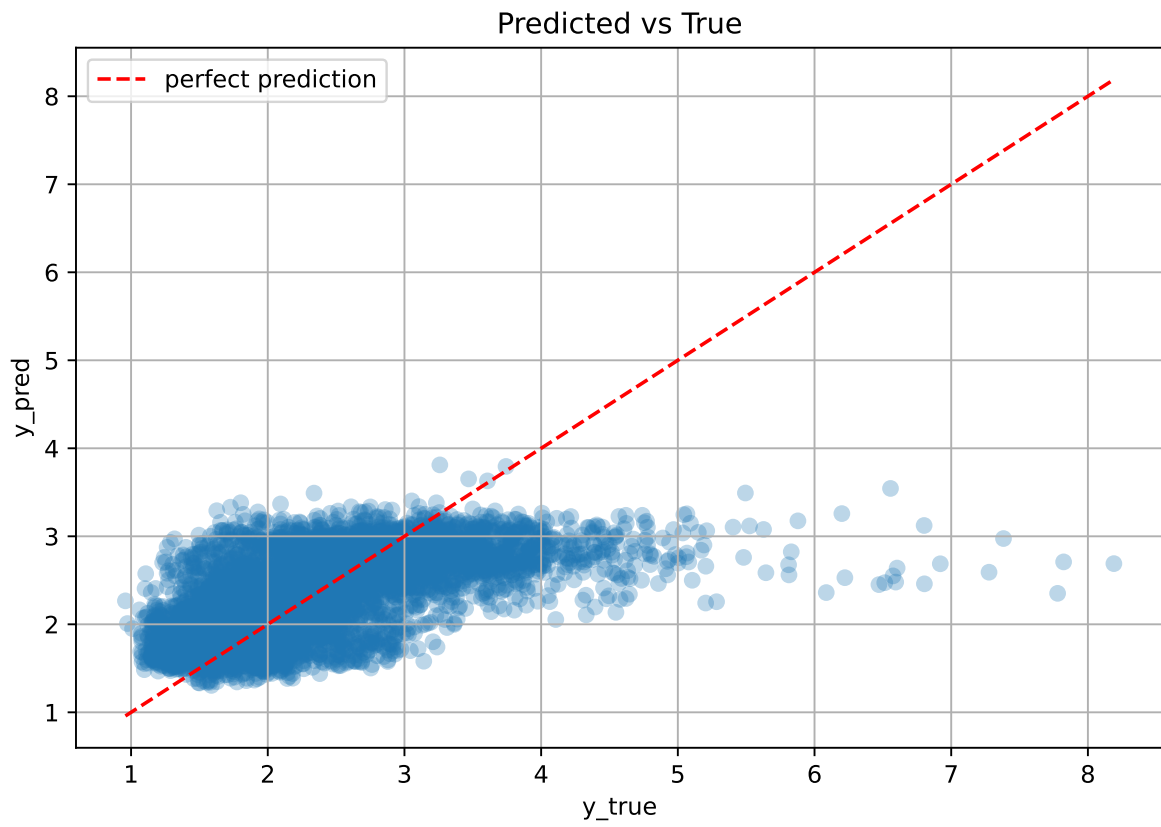
test scores CV: [0.33956739 0.3650984 0.3721525 0.38416477 0.36077412]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.3638	0.3644	0.0147	0.3702	0.3707	0.0063	1.0173	0.9986

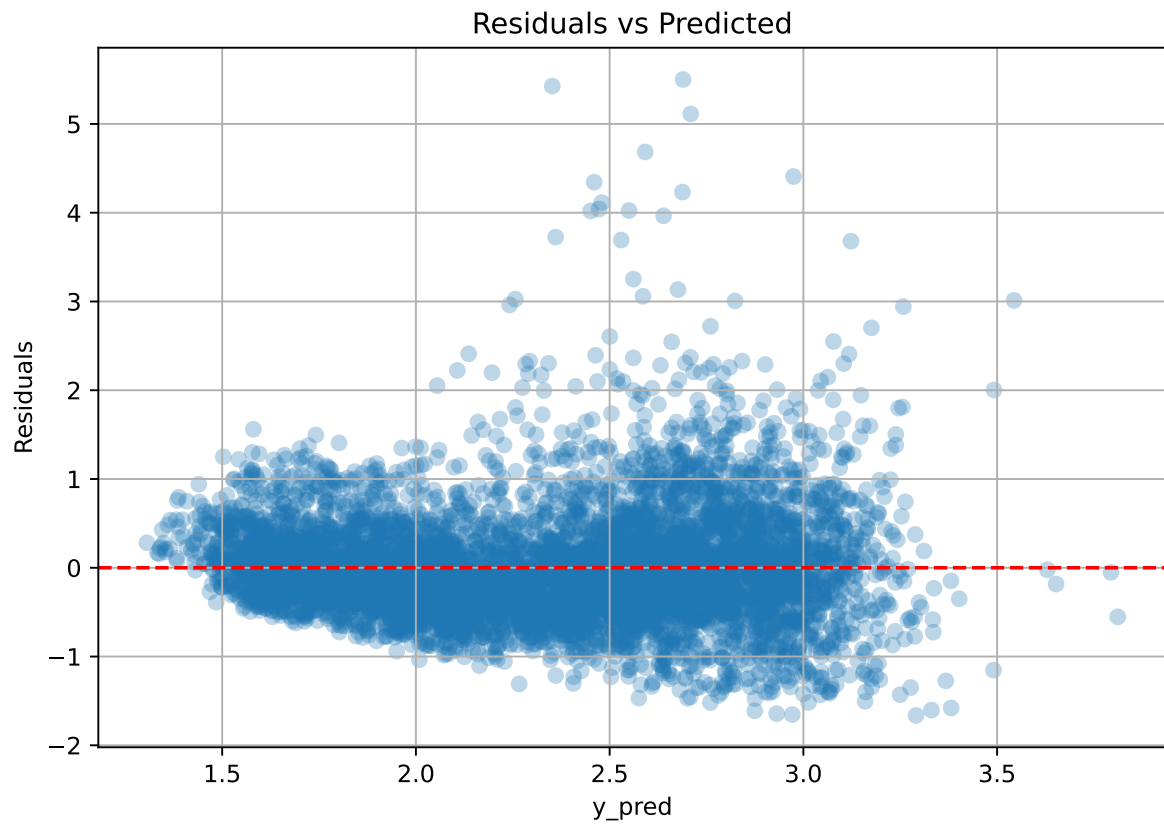
Coefficients:

coef_r_CO2 = -0.0380, coef_r_NH3 = -0.0614, coef_r_NH3_fft_6h_fft_f1_cph = 0.0044, coef_r_NH3_fft_6h_fft_f2_cph = -0.0025, coef_r_NH3_fft_6h_fft_f3_cph = 0.0260, coef_r_NH3_fft_6h_fft_power1 = 0.0553, coef_r_NH3_fft_6h_fft_power3 = -0.0325, coef_r_THI = 0.6218, coef_r_temperature = -0.7418, coef_r_temperature_fft_6h_fft_period1_h = -0.0197, coef_r_temperature_fft_6h_fft_power1 = 0.0308, coef_r_umidity = -0.6045, coef_r_umidity_fft_6h_fft_f1_cph = 0.0170, coef_r_umidity_fft_6h_fft_period2_h = 0.0019, coef_r_umidity_fft_6h_fft_power1 = -0.0574, coef_r_umidity_fft_6h_fft_power2 = 0.0822, intercept = 2.3146,

3.4.2 Scatter



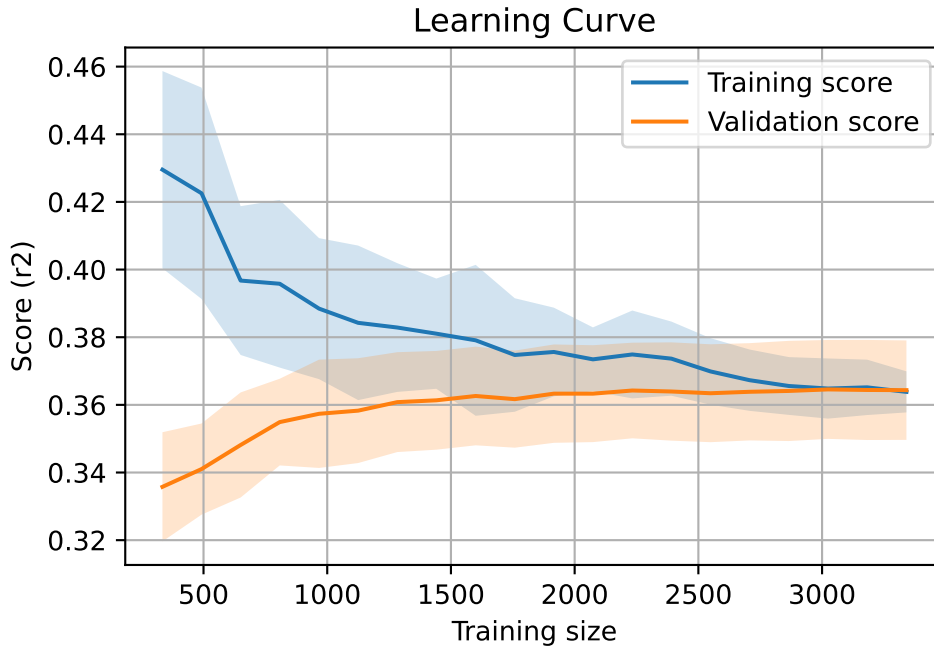
3.4.3 Scatter residuals



3.4.4 Learning curve

`Len(group)` : 6821, `Len(training)` : 4775,

`Len(training_real)` : 3343, `Len(test_of_training)` : 1432, `Len(test)` : 2046



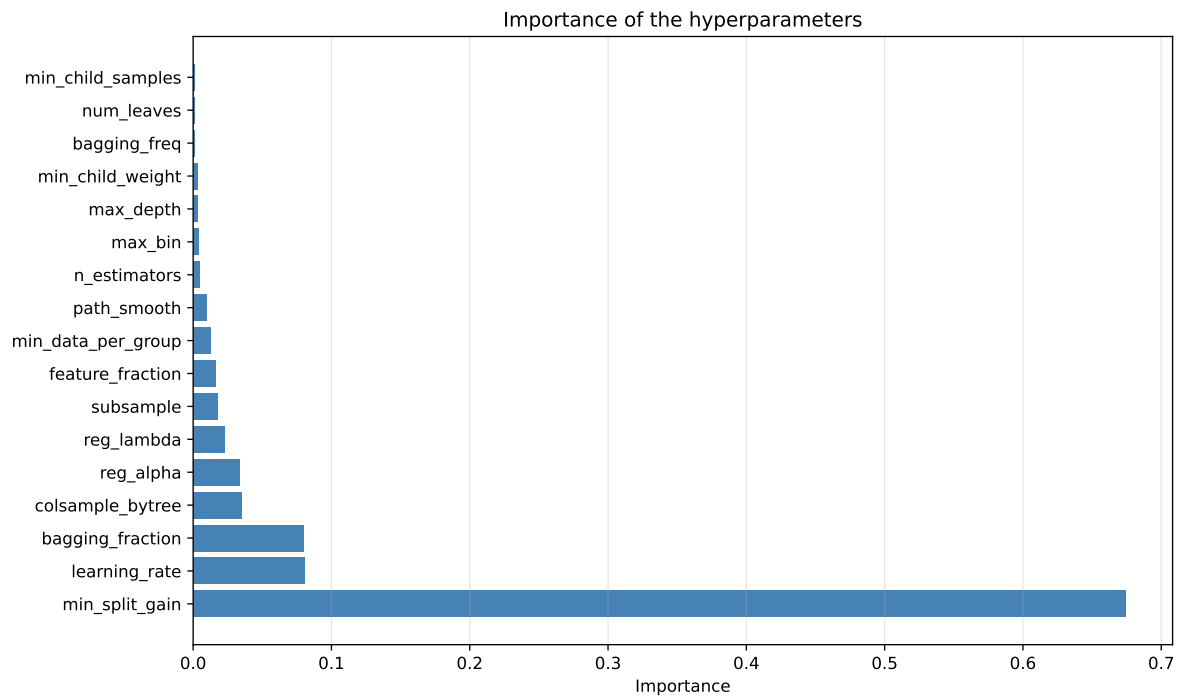
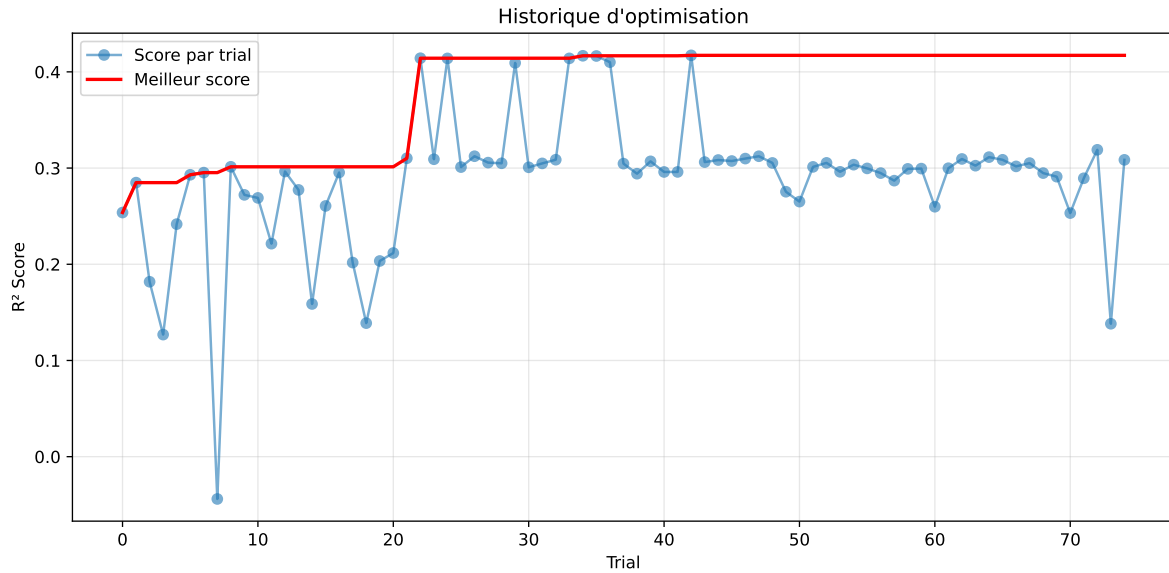
3.5 Model : LGBM

3.5.1 Gridsearch

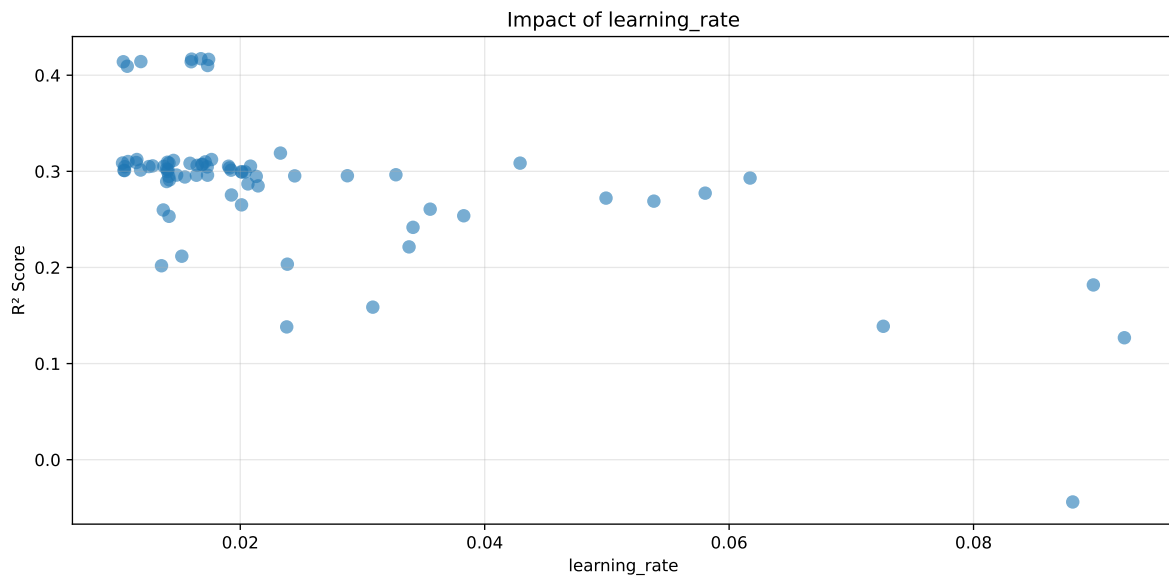
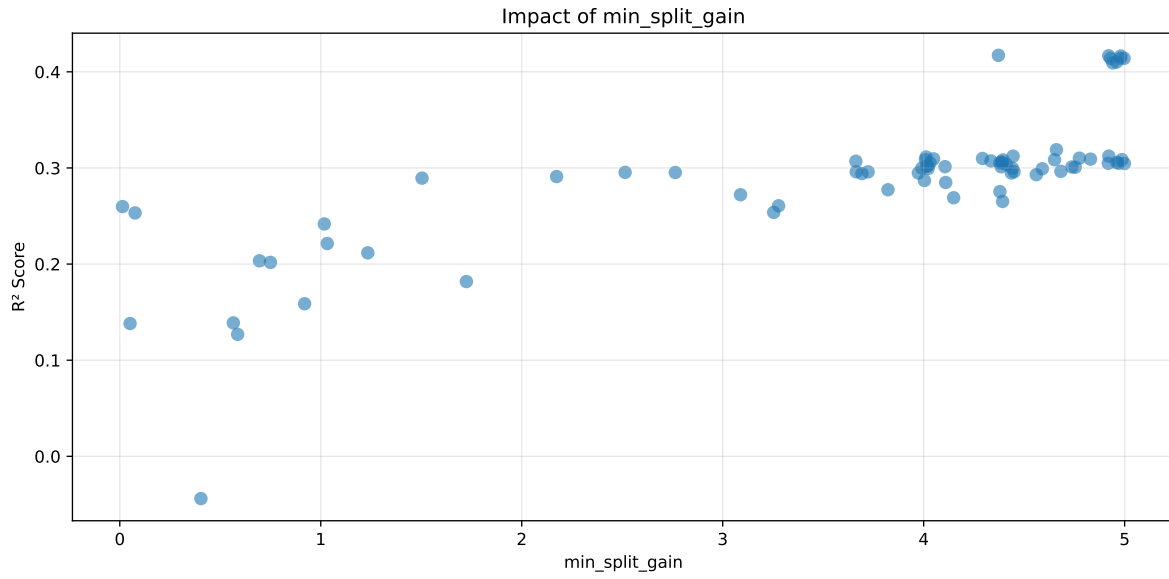
Distribution y_train vs y_test: y_train: mean=2.315, std=0.761, min=0.973, max=7.779
y_test: mean=2.354, std=0.792, min=0.959, max=8.190

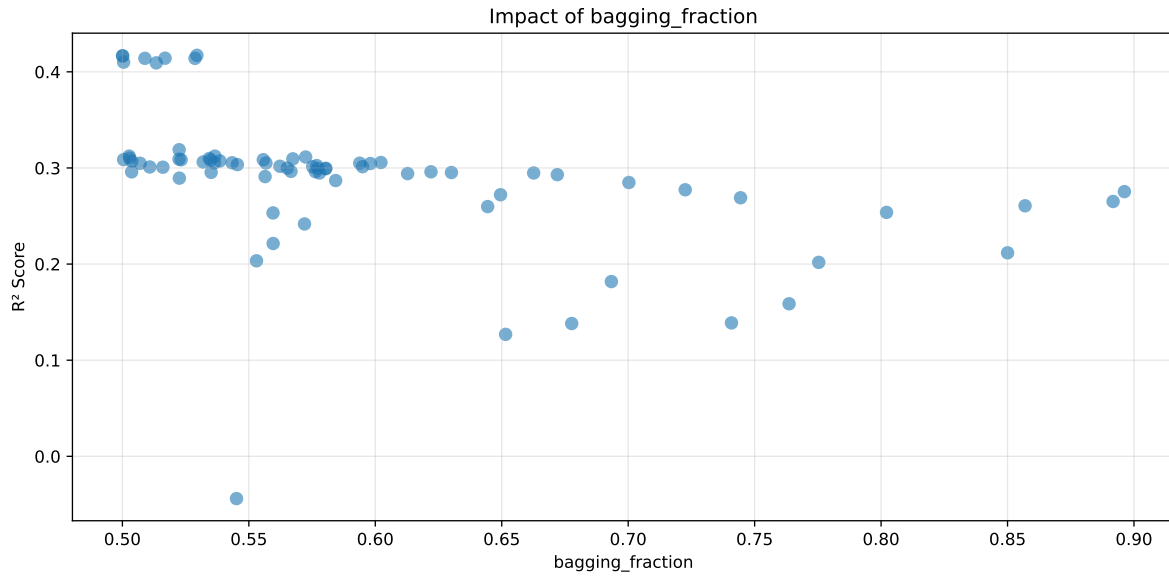
===== TOP Importance FEATURES =====

feature importance	r_umidity	r_NH3	r_temperature	r_CO2	r_THI
214	103	89	87	62	
r_NH3_fft_6h_fft_power1	57	r_temperature_fft_6h_fft_power1	31	r_NH3_fft_6h_fft_power3	
22	r_umidity_fft_6h_fft_power2	16	r_temperature_fft_6h_fft_period1_h	5	r_NH3_fft_6h_fft_f1_cph
4	r_umidity_fft_6h_fft_f1_cph	4	r_NH3_fft_6h_fft_f2_cph	0	r_NH3_fft_6h_fft_f3_cph
0	r_umidity_fft_6h_fft_power1	0	r_umidity_fft_6h_fft_period2_h	0	



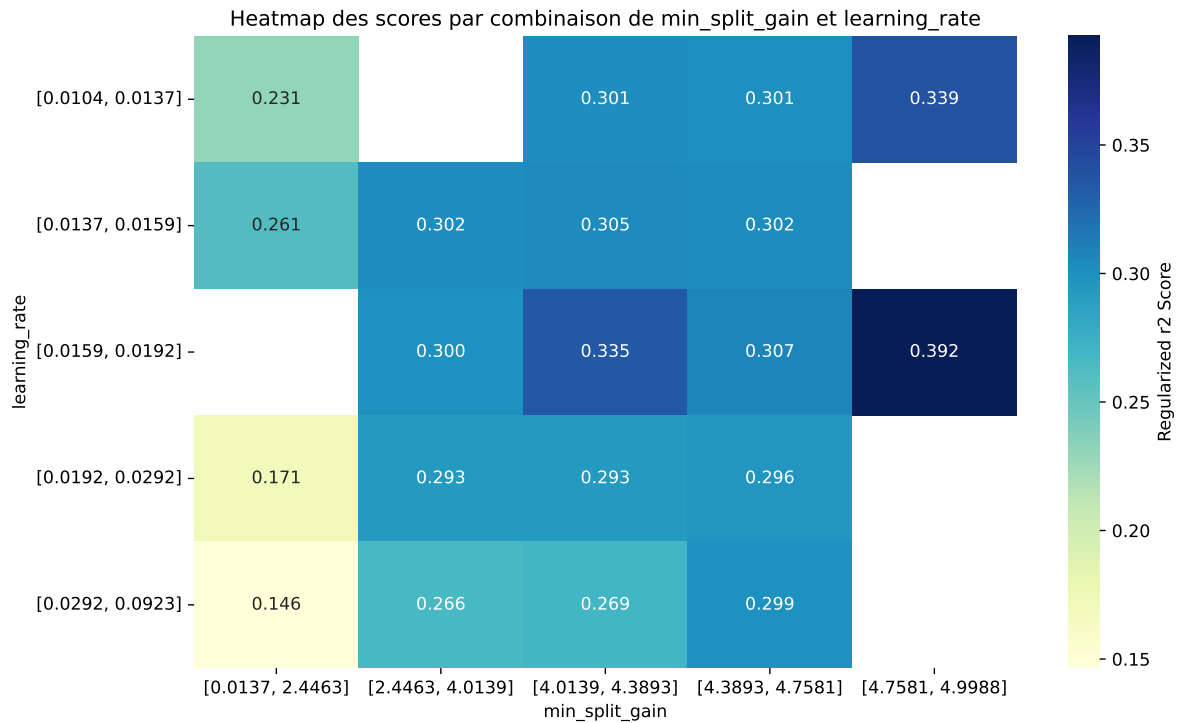
Paramètres avec >5% d'importance : ['min_split_gain', 'learning_rate', 'bagging_fraction']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.4375	0.4172	0.4229	0.0175	0.4172	0.4449	0.0277	1.0665	1.0488
2	0.4353	0.4167	0.4214	0.0173	0.4167	0.4392	0.0224	1.0538	1.0445
3	0.4348	0.4165	0.421	0.0183	0.4165	0.4405	0.024	1.0576	1.0439
4	0.4333	0.4142	0.4193	0.0179	0.4142	0.4391	0.025	1.0603	1.0462
5	0.4316	0.414	0.4181	0.0171	0.414	0.4378	0.0238	1.0574	1.0425
6	0.433	0.414	0.4189	0.0185	0.414	0.4399	0.0259	1.0627	1.0458
7	0.4274	0.4101	0.415	0.0193	0.4101	0.4263	0.0162	1.0395	1.0423
8	0.4277	0.4093	0.4144	0.0173	0.4093	0.4329	0.0236	1.0577	1.0448
9	0.4479	0.4264	0.4306	0.0172	0.319	0.4492	0.0228	1.0535	1.0504
10	0.4383	0.4173	0.4229	0.0178	0.3123	0.4472	0.0299	1.0716	1.0503
---	---	---	---	---	---	---	---	---	---
75	0.697	0.5735	0.5784	0.0198	-0.044	0.6111	0.0376	1.0655	1.2153

Hyperparameters:

Rank 1: {'reg__n_estimators': 200, 'reg__max_depth': 3, 'reg__learning_rate': 0.01677977475135478, 'reg__num_leaves': 21, 'reg__subsample': 0.6196811168005686, 'reg__colsample_bytree': 0.717000431442923, 'reg__min_child_samples': 45, 'reg__reg_alpha': 8.94386605200665, 'reg__reg_lambda': 8.571370395731345, 'reg__min_split_gain': 4.371892233301593, 'reg__path_smooth': 0.6584447639595815, 'reg__min_child_weight': 0.010800846222693844, 'reg__feature_fraction': 0.7351570562538059, 'reg__bagging_fraction': 0.5294586778453144, 'reg__bagging_freq': 3, 'reg__max_bin': 223, 'reg__min_data_per_group': 163}

Rank 2: {'reg__n_estimators': 400, 'reg__max_depth': 3, 'reg__learning_rate': 0.016019209605091383, 'reg__num_leaves': 22, 'reg__subsample': 0.6196761720832078, 'reg__colsample_bytree': 0.6833980413002992, 'reg__min_child_samples': 45, 'reg__reg_alpha': 9.856046272873018, 'reg__reg_lambda': 7.304504063637633, 'reg__min_split_gain': 4.920635665137739, 'reg__path_smooth': 0.5842688256786475, 'reg__min_child_weight': 0.0046917697256131, 'reg__feature_fraction': 0.7429371702653291, 'reg__bagging_fraction': 0.5000527156135187, 'reg__bagging_freq': 1, 'reg__max_bin': 170, 'reg__min_data_per_group': 145}

Rank 3: {'reg__n_estimators': 400, 'reg__max_depth': 3, 'reg__learning_rate': 0.017409227638671077, 'reg__num_leaves': 21, 'reg__subsample': 0.6135448725672441, 'reg__colsample_bytree': 0.7052427816970516, 'reg__min_child_samples': 45, 'reg__reg_alpha': 9.952799056027986, 'reg__reg_lambda': 7.112247999332439, 'reg__min_split_gain': 4.980763160178914, 'reg__path_smooth': 0.5487338470193223, 'reg__min_child_weight': 0.0037189493478354944, 'reg__feature_fraction': 0.7495208042776309, 'reg__bagging_fraction': 0.5001092925545687, 'reg__bagging_freq': 1, 'reg__max_bin': 172, 'reg__min_data_per_group': 151}

Rank 4: {'reg__n_estimators': 400, 'reg__max_depth': 3, 'reg__learning_rate': 0.01186516249280743, 'reg__num_leaves': 18, 'reg__subsample': 0.6646126065754085, 'reg__colsample_bytree': 0.6172071351620658, 'reg__min_child_samples': 50, 'reg__reg_alpha': 9.9029808387743, 'reg__reg_lambda': 0.11551274418903201, 'reg__min_split_gain': 4.980112965820094, 'reg__path_smooth': 0.4473563335877042, 'reg__min_child_weight': 0.0048012931361546, 'reg__feature_fraction': 0.7725630070986267, 'reg__bagging_fraction': 0.5168690584808753, 'reg__bagging_freq': 1, 'reg__max_bin': 199, 'reg__min_data_per_group': 50}

Rank 5: {'reg__n_estimators': 300, 'reg__max_depth': 3, 'reg__learning_rate': 0.015984682121070883, 'reg__num_leaves': 22, 'reg__subsample': 0.6105508761814563, 'reg__colsample_bytree': 0.6953081121013517, 'reg__min_child_samples': 45, 'reg__reg_alpha': 9.777453670218351, 'reg__reg_lambda': 7.182507126100506, 'reg__min_split_gain': 4.997394342986702, 'reg__path_smooth': 0.7711025793219379, 'reg__min_child_weight': 0.00432731408104944, 'reg__feature_fraction': 0.7407327374972101, 'reg__bagging_fraction': 0.5089019283827948, 'reg__bagging_freq': 1, 'reg__max_bin': 214, 'reg__min_data_per_group': 158}

Rank 6: {'reg__n_estimators': 400, 'reg__max_depth': 3, 'reg__learning_rate': 0.01043183173183553, 'reg__num_leaves': 63, 'reg__subsample': 0.6536027163588221, 'reg__colsample_bytree': 0.6203743241057361, 'reg__min_child_samples': 50, 'reg__reg_alpha': 9.965492804358455, 'reg__reg_lambda': 1.2248160768125425, 'reg__min_split_gain': 4.929538758610723, 'reg__path_smooth': 0.284846097017029, 'reg__min_child_weight': 0.004021401785032617, 'reg__feature_fraction': 0.7775343648156746, 'reg__bagging_fraction': 0.5287013215841209, 'reg__bagging_freq': 1, 'reg__max_bin': 191, 'reg__min_data_per_group': 152}

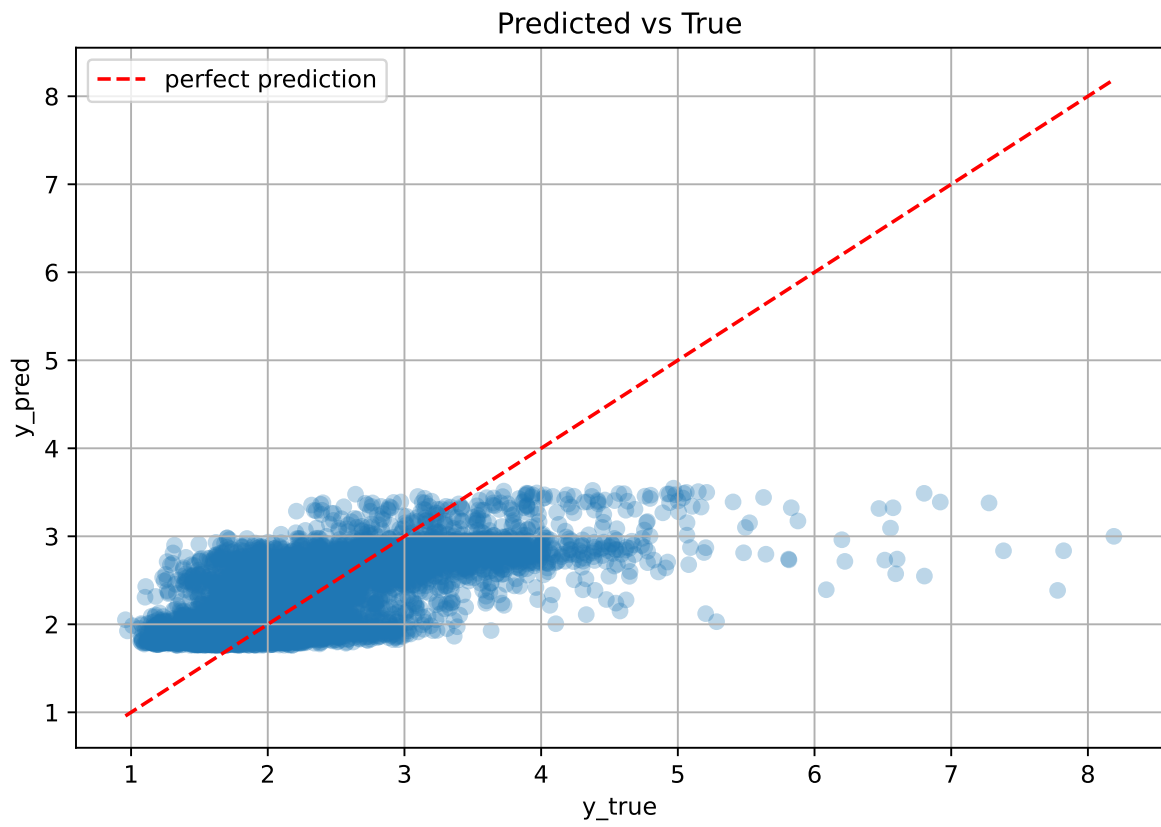
Rank 7: {'reg__n_estimators': 500, 'reg__max_depth': 3, 'reg__learning_rate': 0.017327440923635654, 'reg__num_leaves': 24, 'reg__subsample': 0.6168412389865728, 'reg__colsample_bytree': 0.6824188460644975, 'reg__min_child_samples': 46, 'reg__reg_alpha': 9.913897262725259, 'reg__reg_lambda': 6.801505161457661, 'reg__min_split_gain': 4.959547245027917, 'reg__path_smooth': 0.6837500393221019, 'reg__min_child_weight': 0.006300030528137481, 'reg__feature_fraction': 0.7306766239329062, 'reg__bagging_fraction': 0.5004774891594261, 'reg__bagging_freq': 3, 'reg__max_bin': 177, 'reg__min_data_per_group': 151}

Rank 8: {'reg__n_estimators': 300, 'reg__max_depth': 3, 'reg__learning_rate': 0.010751459388994094, 'reg__num_leaves': 22, 'reg__subsample': 0.779556253446858, 'reg__colsample_bytree': 0.6883270318803426, 'reg__min_child_samples': 45, 'reg__reg_alpha': 9.803801305312152, 'reg__reg_lambda': 0.14013649366420916, 'reg__min_split_gain': 4.9409130595360455, 'reg__path_smooth': 0.5585780847823991, 'reg__min_child_weight': 0.004436495631532782, 'reg__feature_fraction': 0.7321129241578452, 'reg__bagging_fraction': 0.5133971249374876, 'reg__bagging_freq': 1, 'reg__max_bin': 190, 'reg__min_data_per_group': 75}

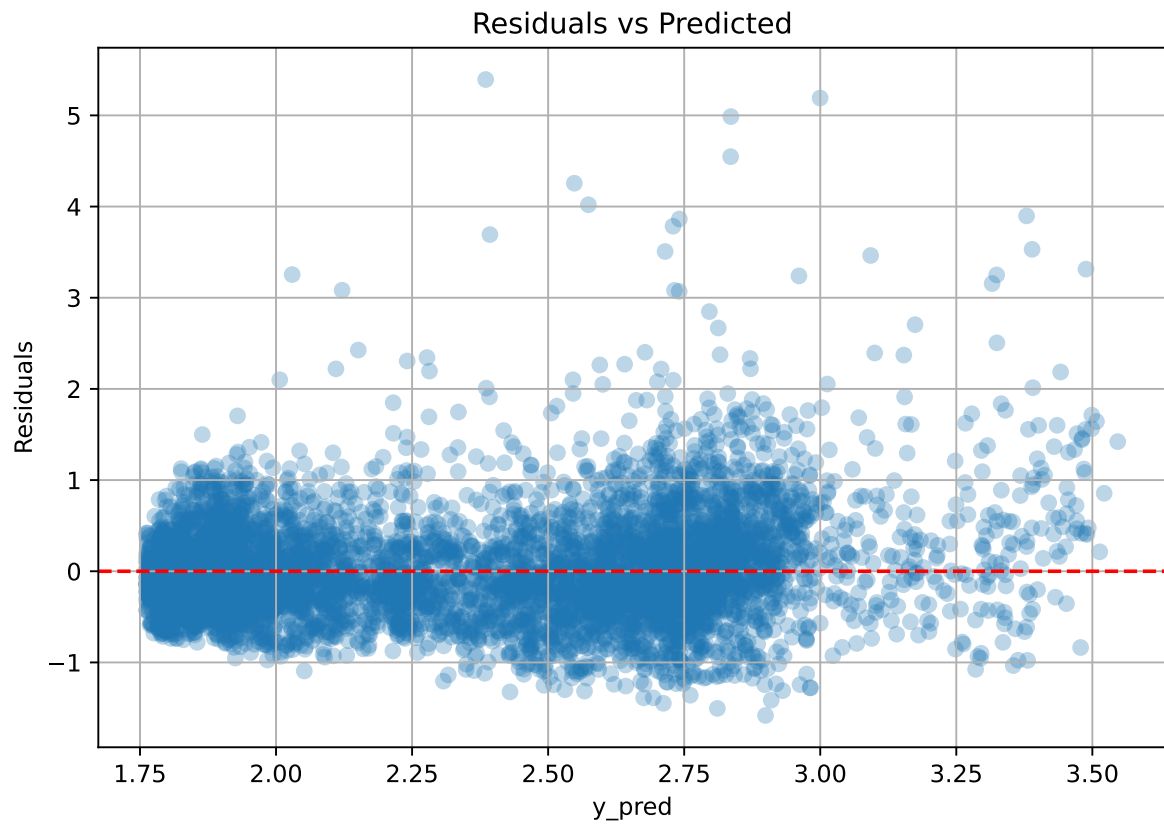
Rank 9: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.023287347616757575, 'reg__num_leaves': 26, 'reg__subsample': 0.5623334470491058, 'reg__colsample_bytree': 0.6224735234457719, 'reg__min_child_samples': 49, 'reg__reg_alpha': 9.454665033035035, 'reg__reg_lambda': 5.682567132282389, 'reg__min_split_gain': 4.660812357603943, 'reg__path_smooth': 0.4150237515983376, 'reg__min_child_weight': 0.033400685238666905, 'reg__feature_fraction': 0.7510828561297396, 'reg__bagging_fraction': 0.5224461699460687, 'reg__bagging_freq': 1, 'reg__max_bin': 157, 'reg__min_data_per_group': 155}

Rank 10: {'reg__n_estimators': 400, 'reg__max_depth': 7, 'reg__learning_rate': 0.011537665294640368, 'reg__num_leaves': 63, 'reg__subsample': 0.6501789744163127, 'reg__colsample_bytree': 0.6451062667783355, 'reg__min_child_samples': 33, 'reg__reg_alpha': 9.801752300588172, 'reg__reg_lambda': 3.256882919899004, 'reg__min_split_gain': 4.92122428991357, 'reg__path_smooth': 0.032669194851953964, 'reg__min_child_weight': 0.0051124277254837504, 'reg__feature_fraction': 0.7727849030245417, 'reg__bagging_fraction': 0.5026714800611854, 'reg__bagging_freq': 1, 'reg__max_bin': 195, 'reg__min_data_per_group': 73}

3.5.2 Scatter



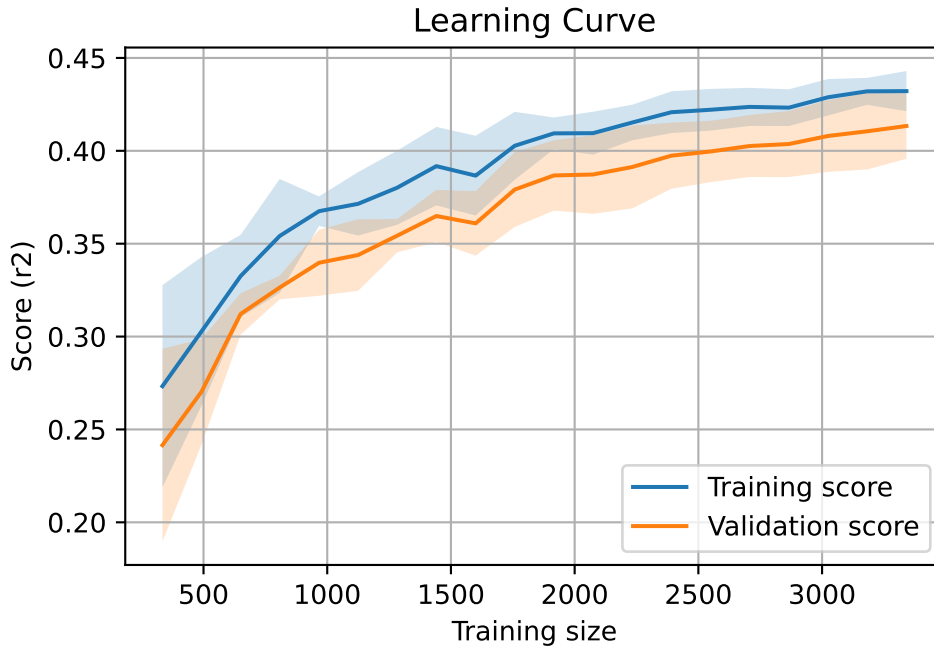
3.5.3 Scatter residuals



3.5.4 Learning curve

`Len(group)` : 6821, `Len(training)` : 4775,

`Len(training_real)` : 3343, `Len(test_of_training)` : 1432, `Len(test)` : 2046



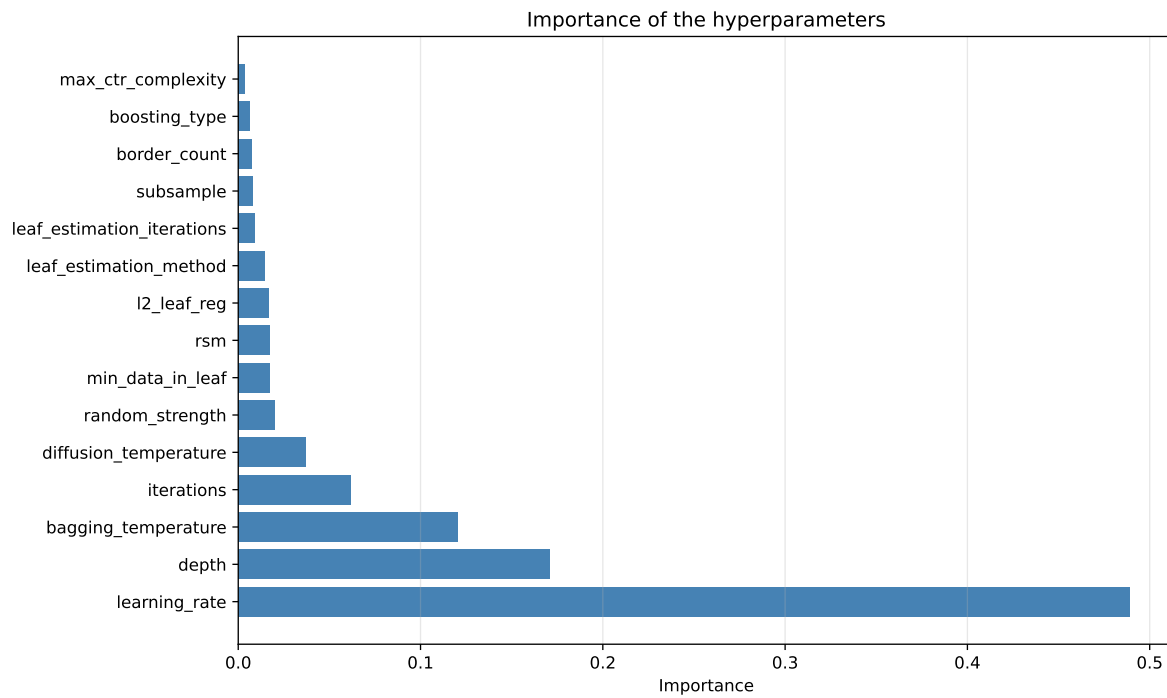
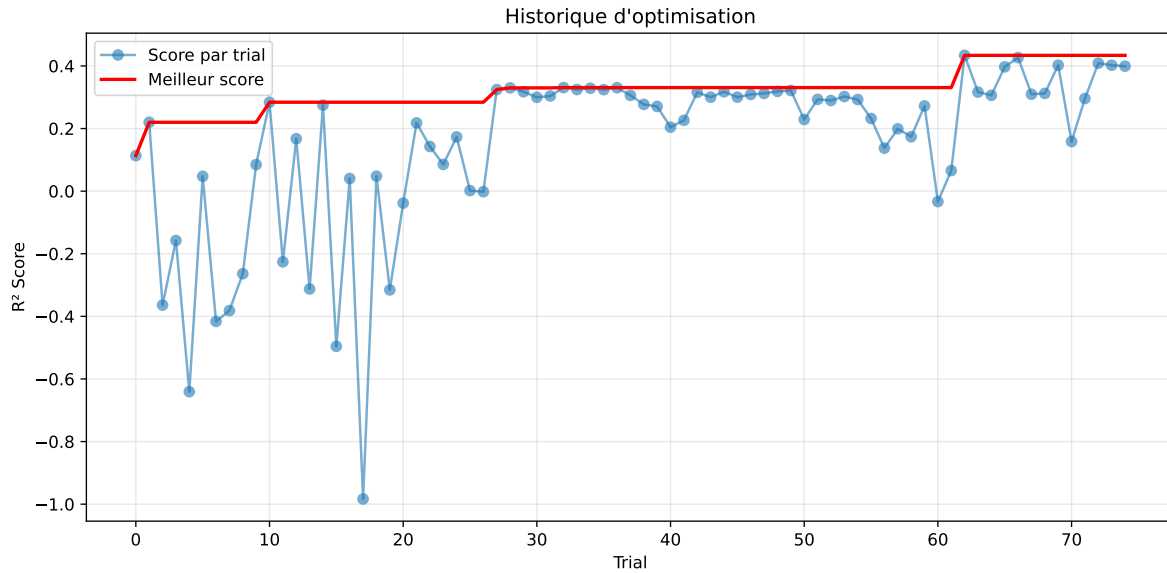
3.6 Model : CatBoost

3.6.1 Gridsearch

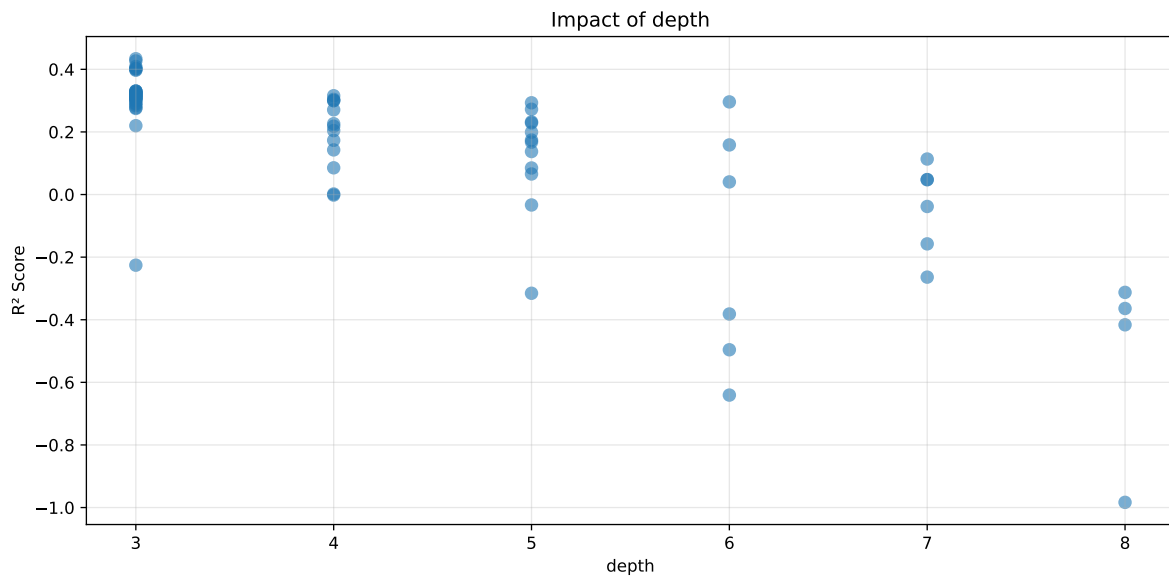
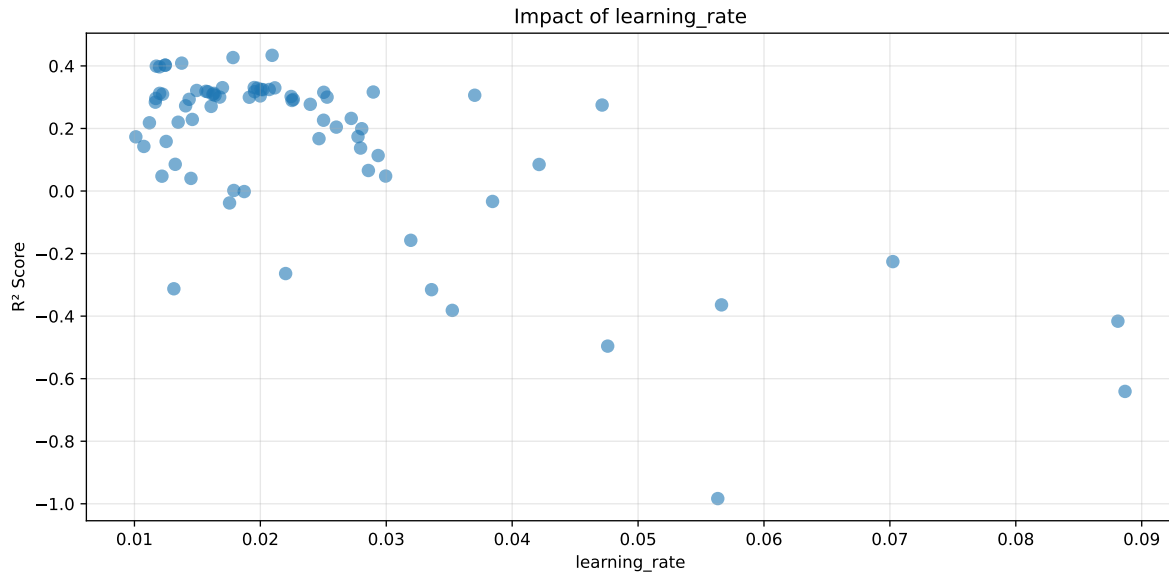
Distribution y_train vs y_test: y_train: mean=2.315, std=0.761, min=0.973, max=7.779
y_test: mean=2.354, std=0.792, min=0.959, max=8.190

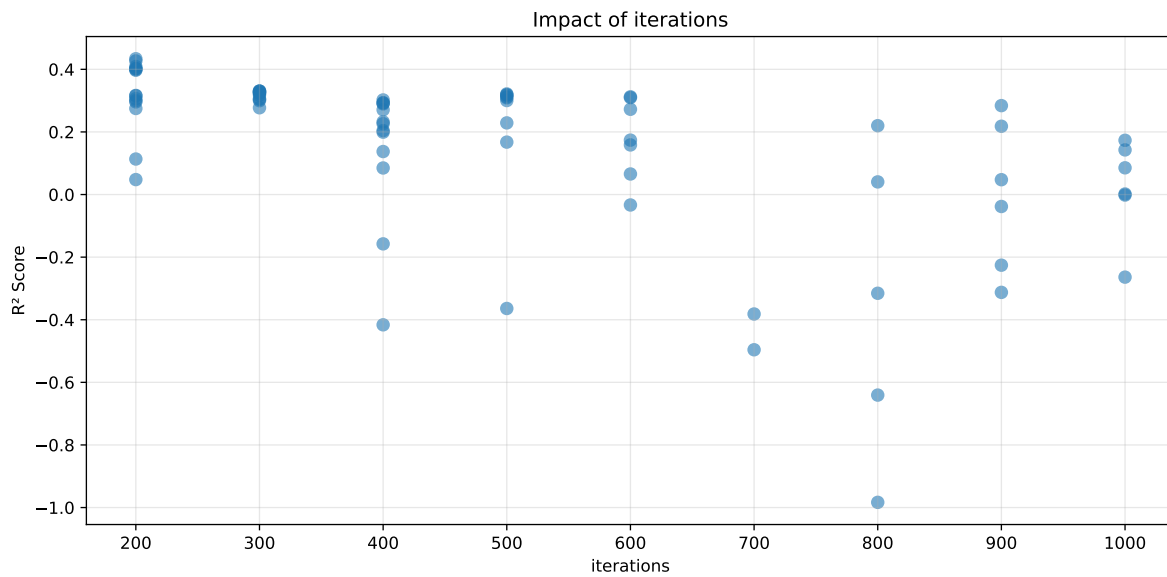
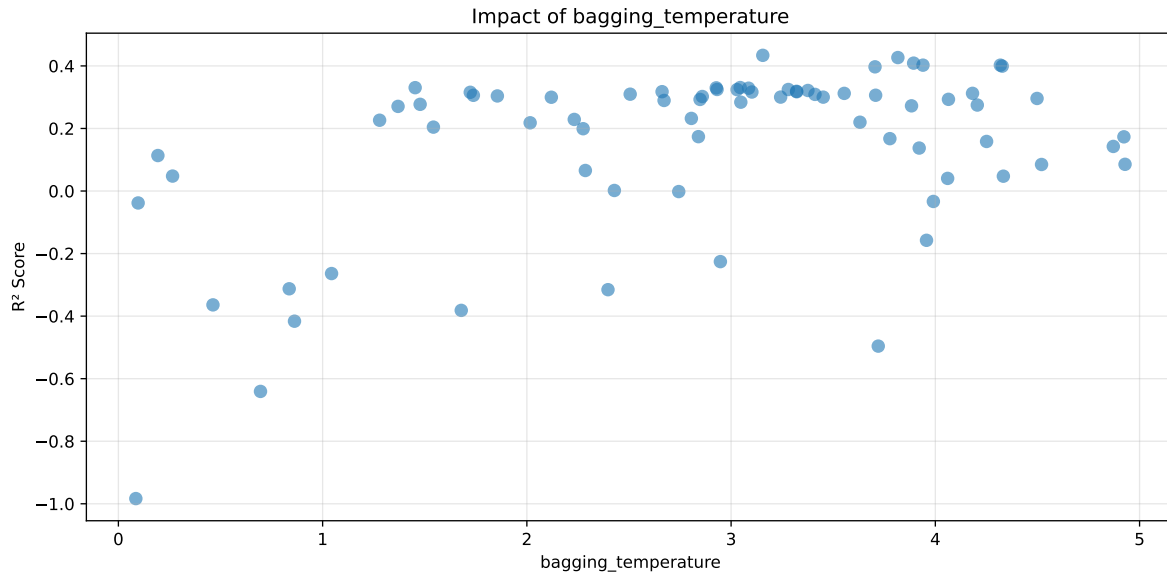
===== TOP Importance FEATURES =====

feature importance	feature	importance
53.593469	r_umidity	53.593469
13.949986	r_NH3	13.949986
11.198615	r_temperature	11.198615
5.442576	r_THI	4.671242
4.671242	r_NH3_fft_6h_fft_power1	3.030616
3.030616	r_NH3_fft_6h_fft_power3	1.939297
1.939297	r_umidity_fft_6h_fft_power2	1.593678
1.593678	r_temperature_fft_6h_fft_power1	1.488855
1.488855	r_temperature_fft_6h_fft_period1_h	1.183880
1.183880	r_umidity_fft_6h_fft_power1	0.443239
0.443239	r_umidity_fft_6h_fft_f1_cph	0.411945
0.411945	r_umidity_fft_6h_fft_period2_h	0.398251
0.398251	r_NH3_fft_6h_fft_f1_cph	0.371398
0.371398	r_NH3_fft_6h_fft_f3_cph	0.152976
0.152976	r_NH3_fft_6h_fft_f2_cph	0.129976



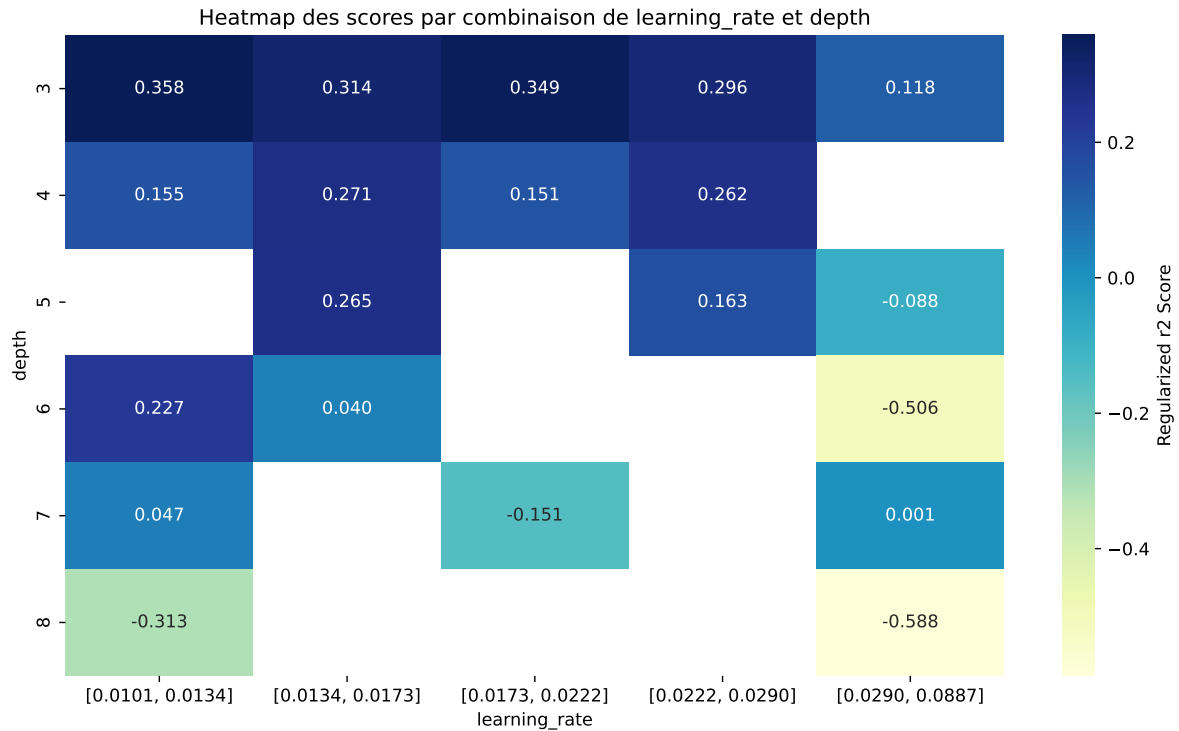
Paramètres avec >5% d'importance : ['learning_rate', 'depth', 'bagging_temperature', 'iterations']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.4527	0.4338	0.4349	0.018	0.4338	0.4404	0.0065	1.015	1.0434
2	0.445	0.4267	0.4291	0.0182	0.4267	0.4321	0.0053	1.0125	1.0429
3	0.4206	0.4089	0.4095	0.0164	0.4089	0.4111	0.0022	1.0055	1.0286
4	0.4138	0.4024	0.4015	0.0146	0.4024	0.4013	-0.0011	0.9972	1.0284
5	0.4133	0.4024	0.4011	0.0161	0.4024	0.4048	0.0024	1.006	1.0271
6	0.4104	0.3993	0.3981	0.0156	0.3993	0.3988	-0.0005	0.9987	1.0277
7	0.4062	0.3971	0.3969	0.0151	0.3971	0.4002	0.0031	1.0078	1.0229
8	0.4892	0.4628	0.4632	0.0169	0.3309	0.4652	0.0024	1.0051	1.057
9	0.4754	0.4512	0.4533	0.0162	0.3305	0.459	0.0077	1.0171	1.0535
10	0.4817	0.4564	0.4602	0.0175	0.3299	0.4667	0.0103	1.0226	1.0554
---	---	---	---	---	---	---	---	---	---
75	0.9801	0.6529	0.6613	0.0132	-0.9834	0.6858	0.0329	1.0504	1.5012

Hyperparameters:

Rank 1: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.02093530602341605, 'reg__l2_leaf_reg': 46.93295248856588, 'reg__bagging_temperature': 3.155169572561437, 'reg__random_strength': 1.7830759729142818, 'reg__subsample': 0.5643599047930419, 'reg__min_data_in_leaf': 18, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.5845796484221054, 'reg__border_count': 58, 'reg__leaf_estimation_method': 'Newton',

'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 4069.1498363552573}

Rank 2: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.01782743725362256, 'reg__l2_leaf_reg': 46.32516620041092, 'reg__bagging_temperature': 3.8165913986957003, 'reg__random_strength': 1.6315749036150609, 'reg__subsample': 0.5390717080347271, 'reg__min_data_in_leaf': 17, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.5757136151211757, 'reg__border_count': 81, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 4109.317835259693}

Rank 3: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.013752563185007548, 'reg__l2_leaf_reg': 42.87483069581435, 'reg__bagging_temperature': 3.8937126417779844, 'reg__random_strength': 1.74290122893375, 'reg__subsample': 0.5283939317273434, 'reg__min_data_in_leaf': 34, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.5809897019349423, 'reg__border_count': 83, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 2593.8632329293496}

Rank 4: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.012435523503520183, 'reg__l2_leaf_reg': 38.24164463635211, 'reg__bagging_temperature': 3.9395040989249974, 'reg__random_strength': 1.7556038612590057, 'reg__subsample': 0.5896306777626357, 'reg__min_data_in_leaf': 18, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.520933495112322, 'reg__border_count': 80, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 3364.445283499731}

Rank 5: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.01245489172993697, 'reg__l2_leaf_reg': 36.759377363187156, 'reg__bagging_temperature': 4.318370981062331, 'reg__random_strength': 1.7962202797928124, 'reg__subsample': 0.5317272123324673, 'reg__min_data_in_leaf': 18, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.5172278554218094, 'reg__border_count': 82, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 3095.642272109176}

Rank 6: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.011734162315363561, 'reg__l2_leaf_reg': 38.72610427190748, 'reg__bagging_temperature': 4.327564873490302, 'reg__random_strength': 1.691450185543527, 'reg__subsample': 0.5000875571974206, 'reg__min_data_in_leaf': 36, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.5223273211229825, 'reg__border_count': 83, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 2601.948950491543}

Rank 7: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.011988629269515577, 'reg__l2_leaf_reg': 46.818600755058085, 'reg__bagging_temperature': 3.7046854836930105, 'reg__random_strength': 1.7687718092676044, 'reg__subsample': 0.5300141960397164,

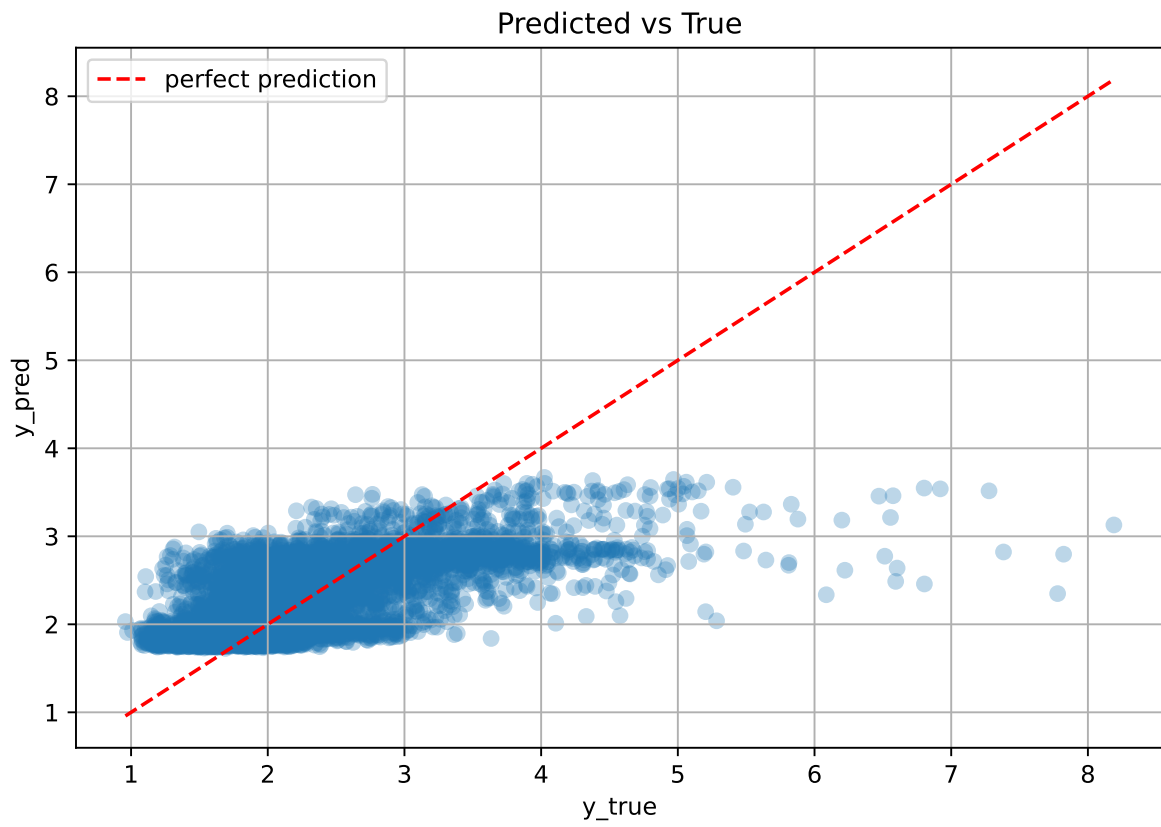
'reg__min_data_in_leaf': 13, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.5795814079732592, 'reg__border_count': 58, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 4134.881836938133}

Rank 8: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.019512262679050763, 'reg__l2_leaf_reg': 48.93494588334056, 'reg__bagging_temperature': 3.044687132115481, 'reg__random_strength': 2.037967235961556, 'reg__subsample': 0.6616647429556292, 'reg__min_data_in_leaf': 24, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.6252517933237906, 'reg__border_count': 67, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 5815.1892493851265}

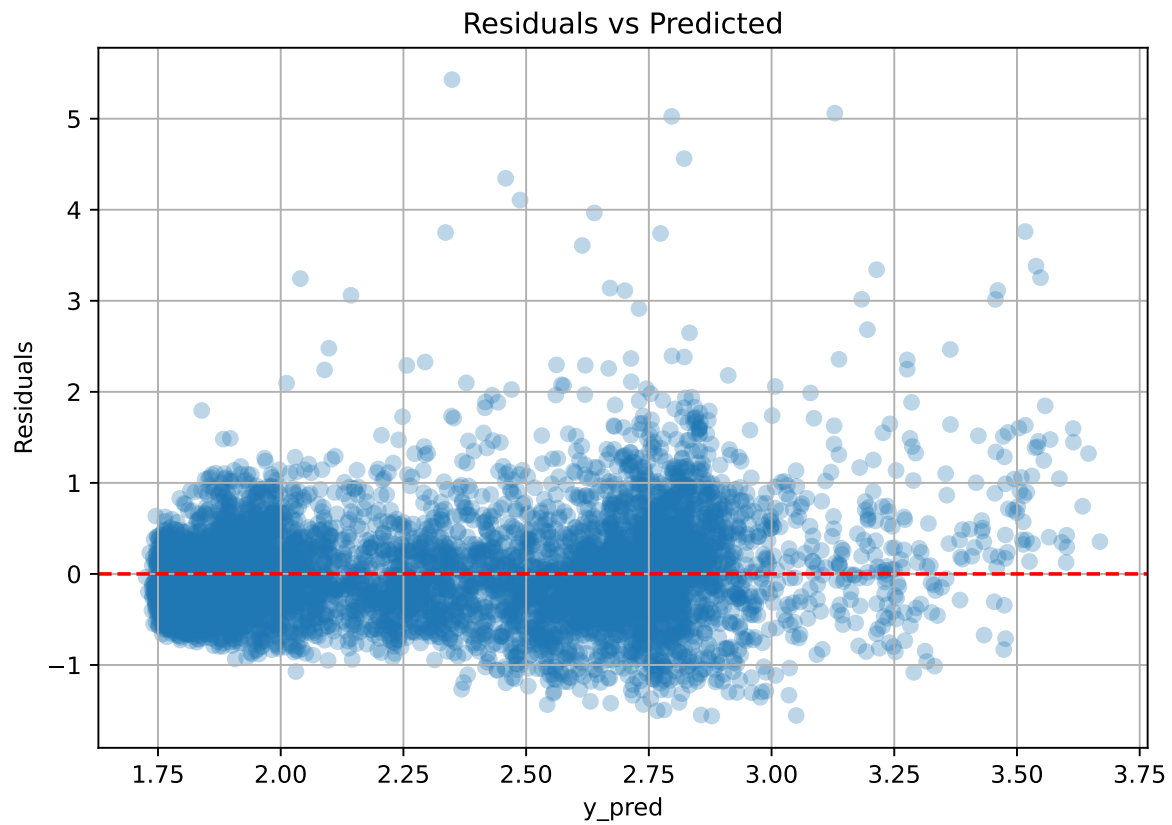
Rank 9: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.01698963125447995, 'reg__l2_leaf_reg': 44.41041048791257, 'reg__bagging_temperature': 1.4530919608182493, 'reg__random_strength': 1.9976075523437626, 'reg__subsample': 0.6541670500580862, 'reg__min_data_in_leaf': 15, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.6512832539342188, 'reg__border_count': 74, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 5763.198835223772}

Rank 10: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.021148370188797314, 'reg__l2_leaf_reg': 49.67317761074678, 'reg__bagging_temperature': 2.9273049423657356, 'reg__random_strength': 2.0582542876082615, 'reg__subsample': 0.7116529664166371, 'reg__min_data_in_leaf': 18, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.602870332329664, 'reg__border_count': 68, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 5568.50330983931}

3.6.2 Scatter



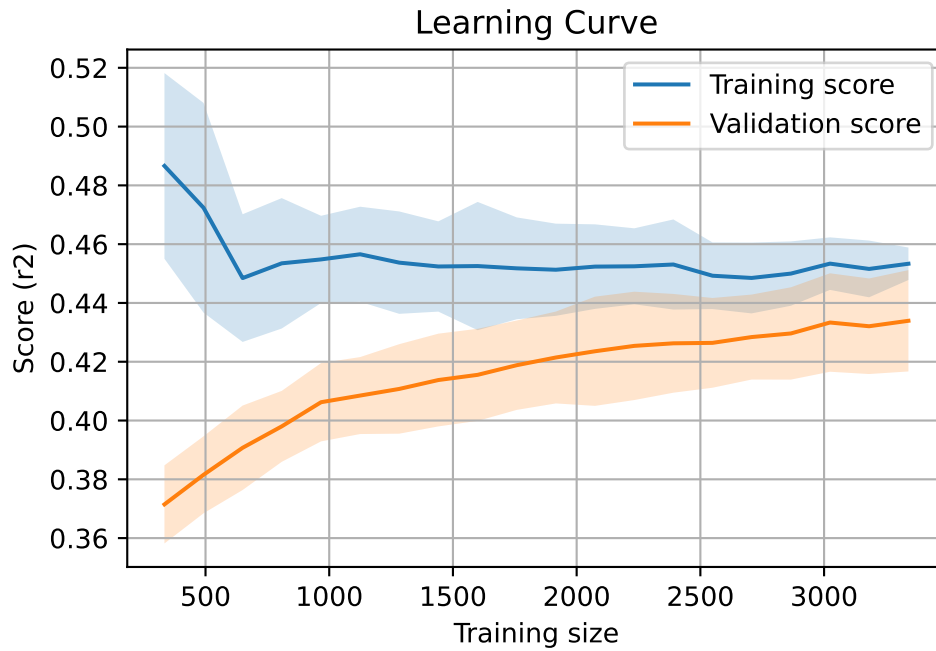
3.6.3 Scatter residuals



3.6.4 Learning curve

`Len(group)` : 6821, `Len(training)` : 4775,

`Len(training_real)` : 3343, `Len(test_of_training)` : 1432, `Len(test)` : 2046



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.364	0.364	0.015	0.37	0.371	0.006	1.017	0.999	nan
LGBM	0.438	0.417	0.018	0.423	0.445	0.028	1.066	1.049	0.4172
CatBoost	0.453	0.434	0.018	0.435	0.44	0.006	1.015	1.043	0.4338

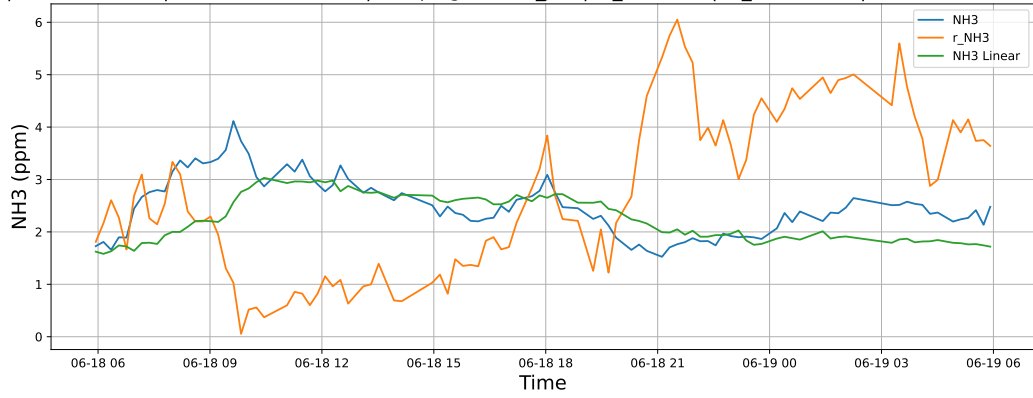
4 Plot

4.1 Standalone plots

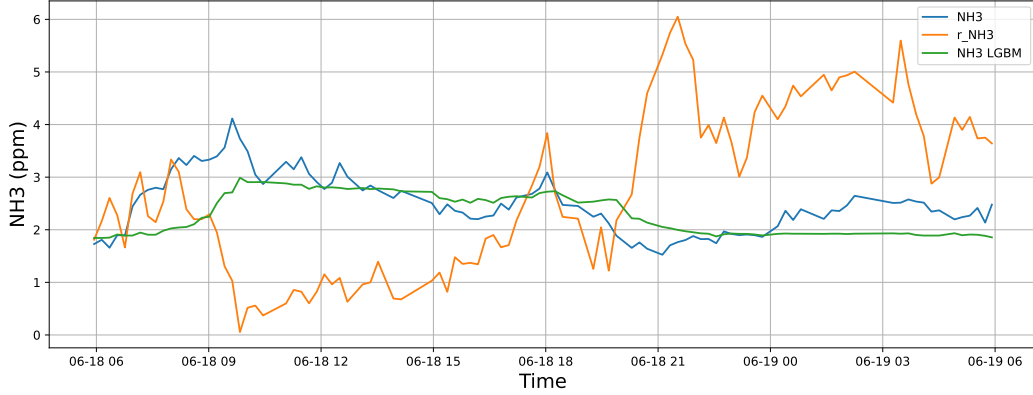
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

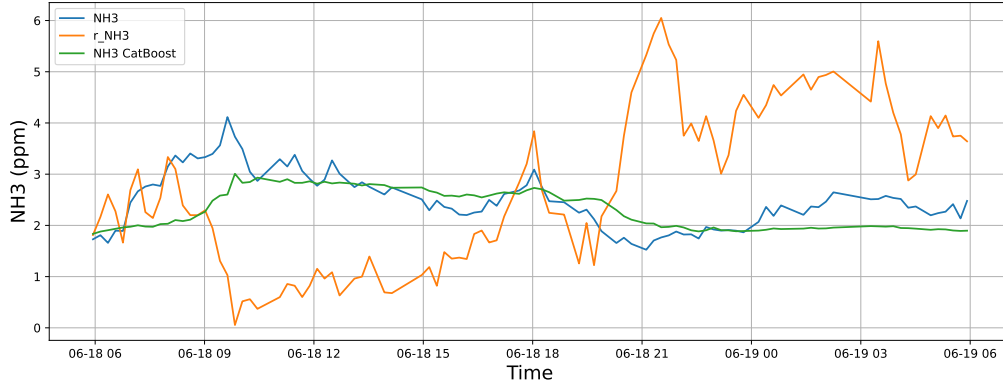
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

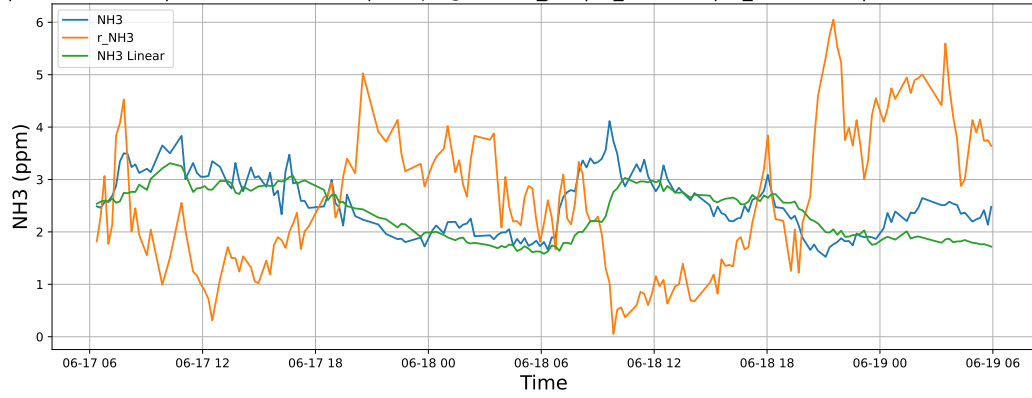


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

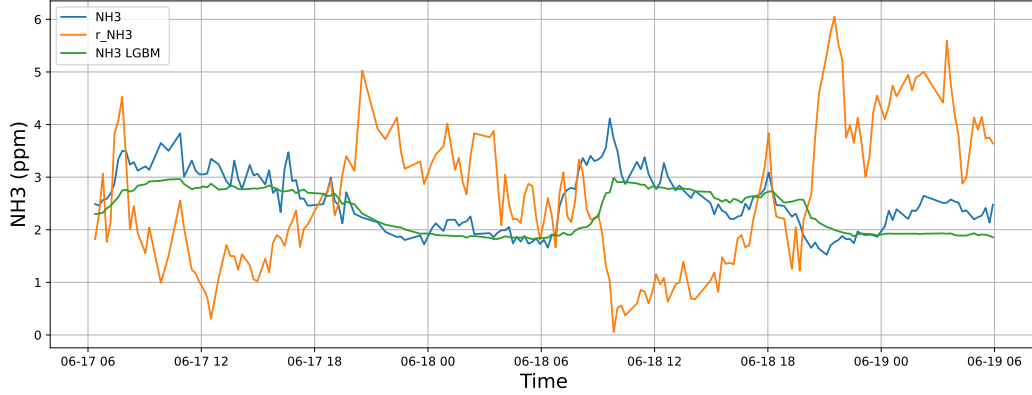


4.1.1.2 Time window : 48

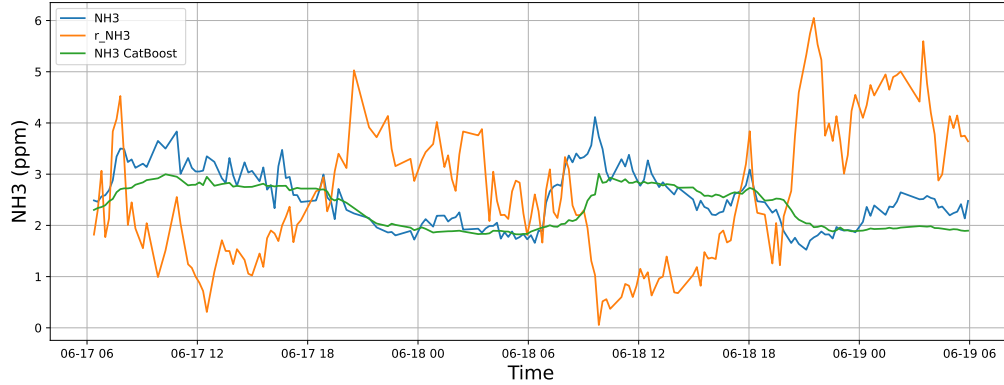
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

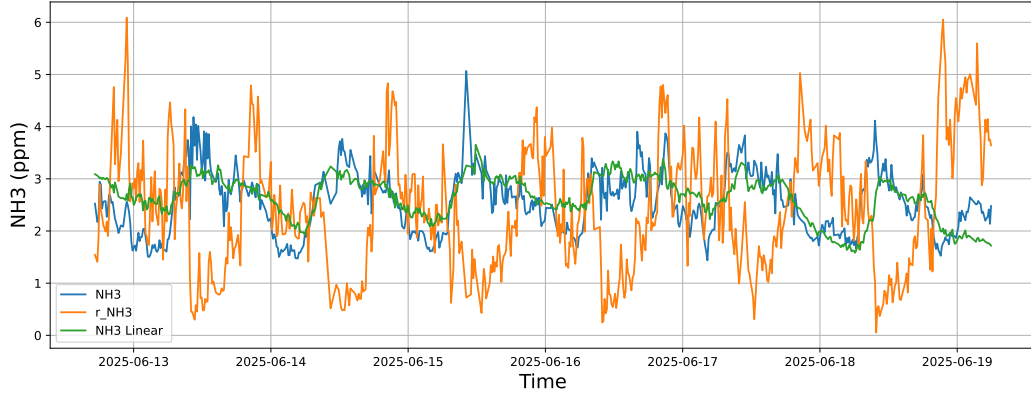


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

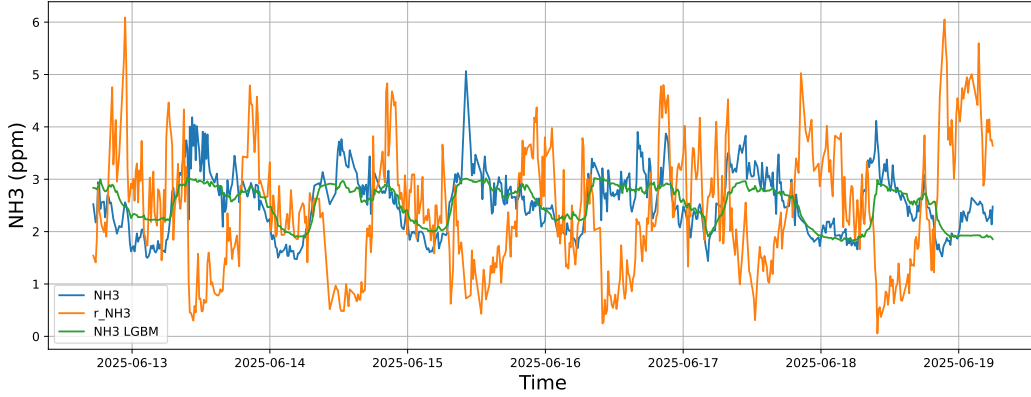


4.1.1.3 Time window : ALL

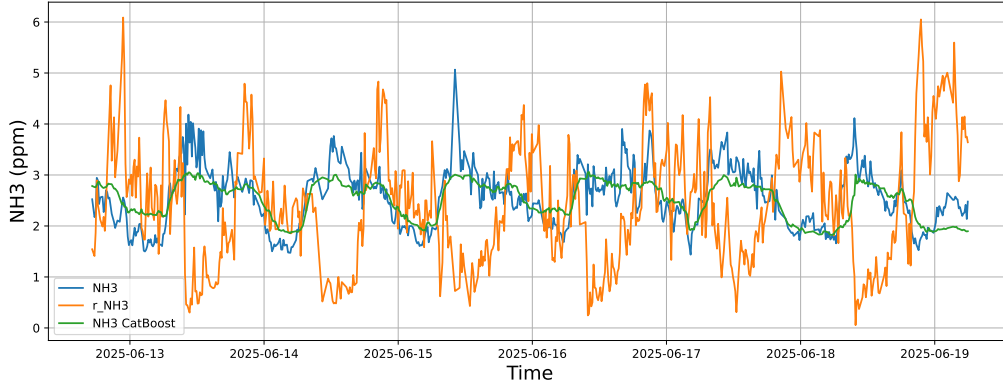
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



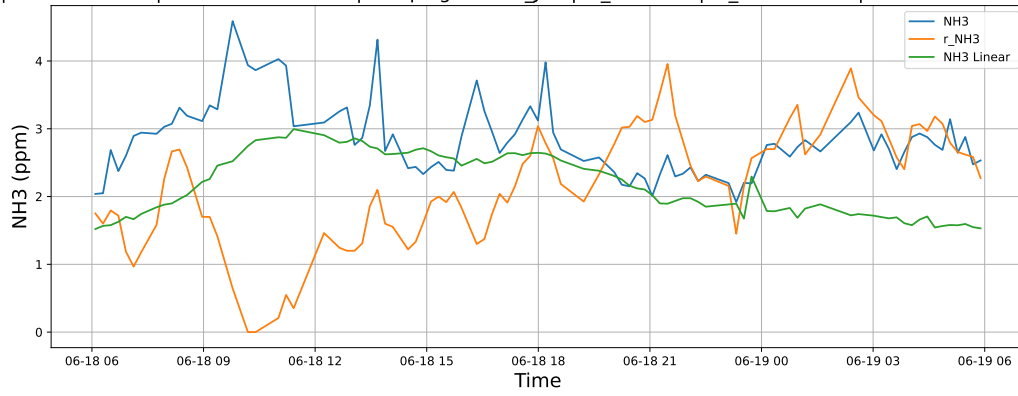
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



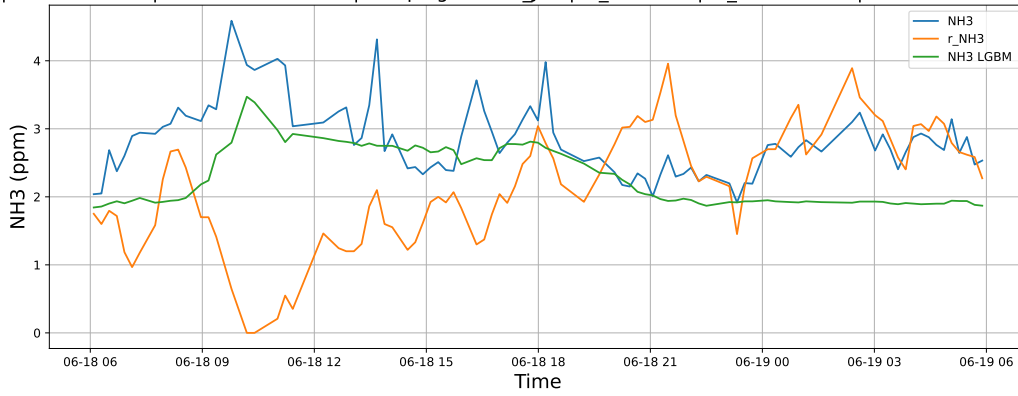
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

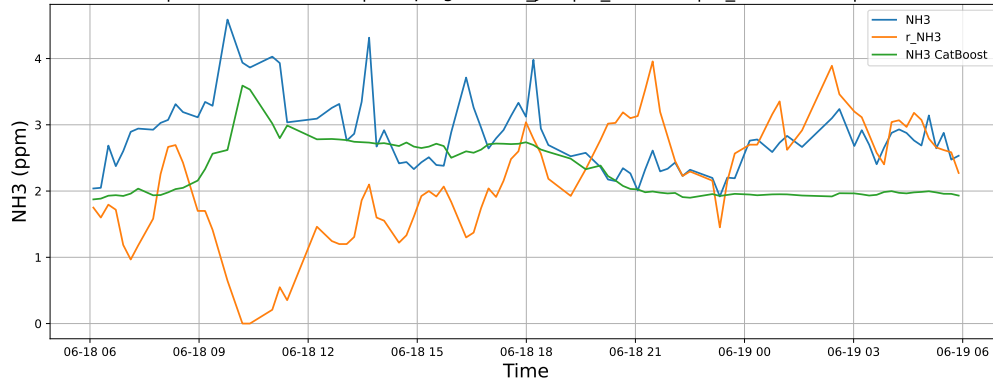
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

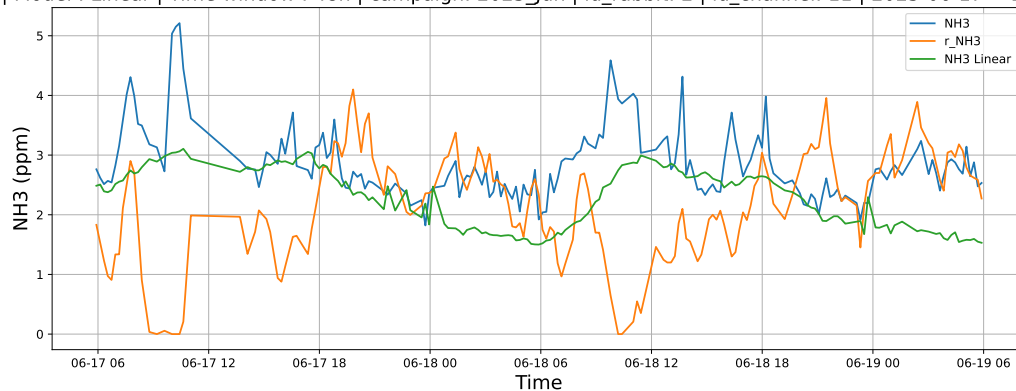


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

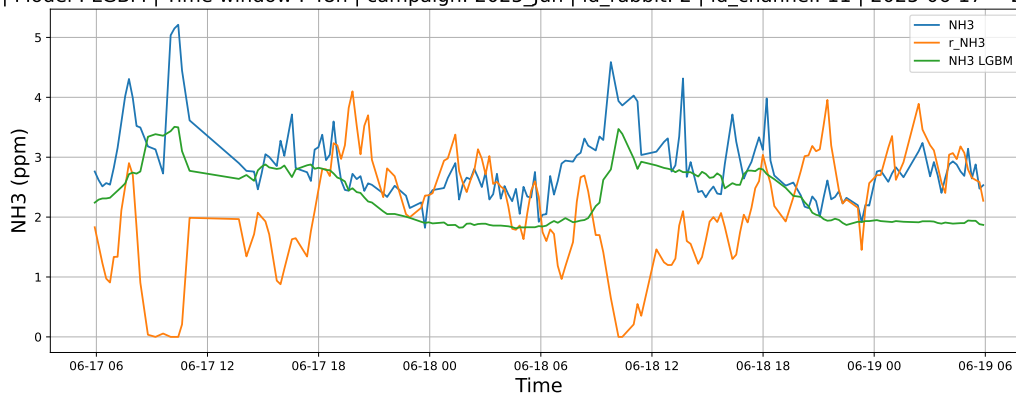


4.1.2.2 Time window : 48

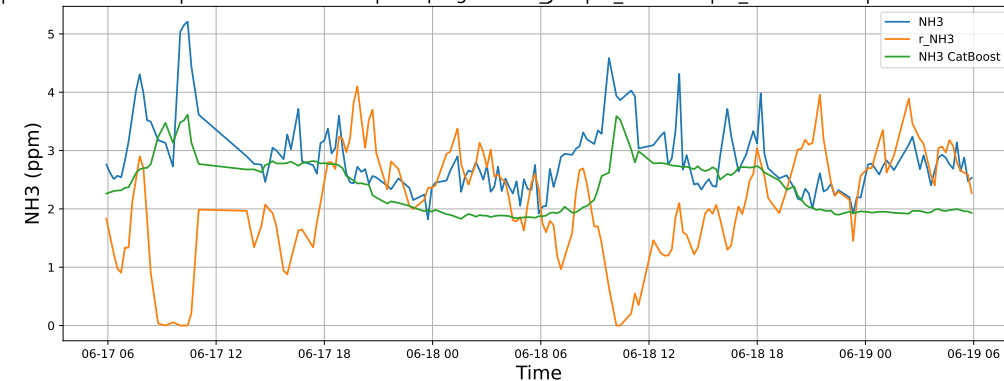
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

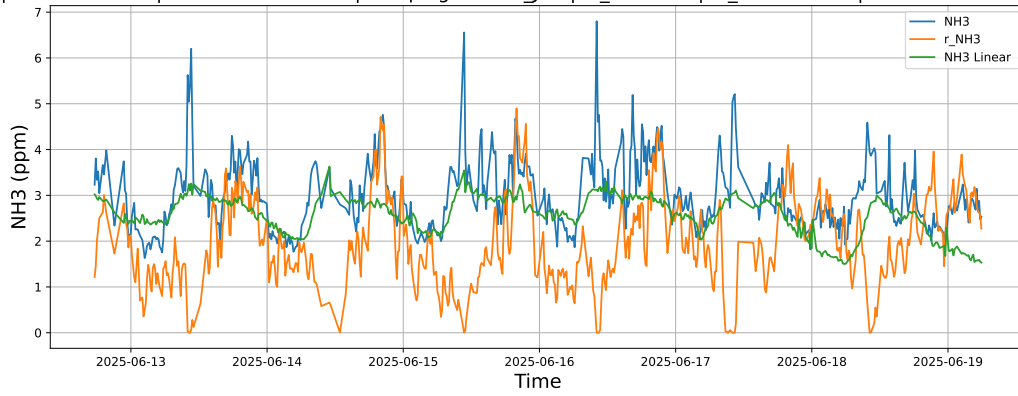


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

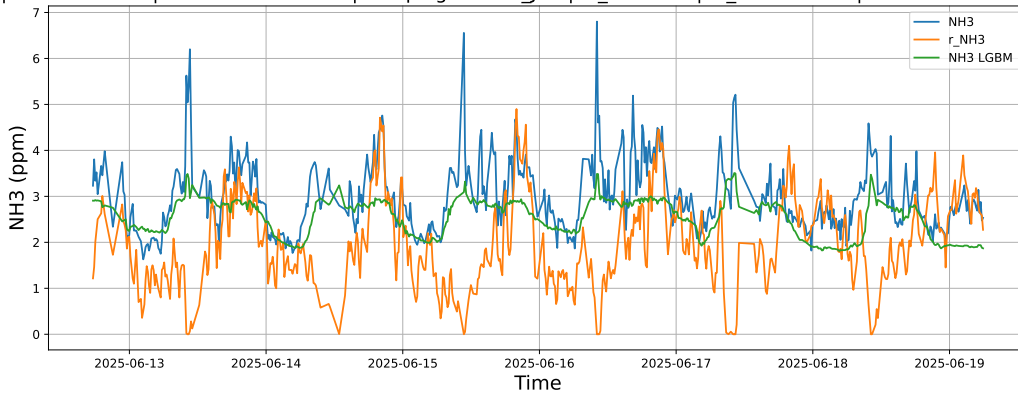


4.1.2.3 Time window : ALL

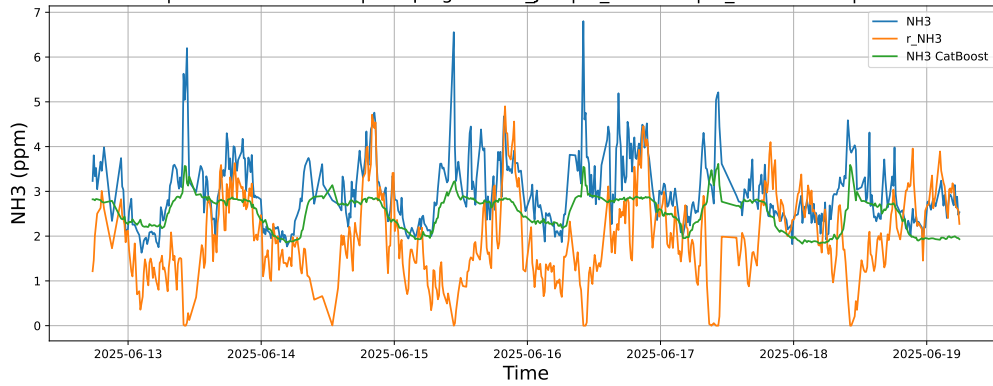
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



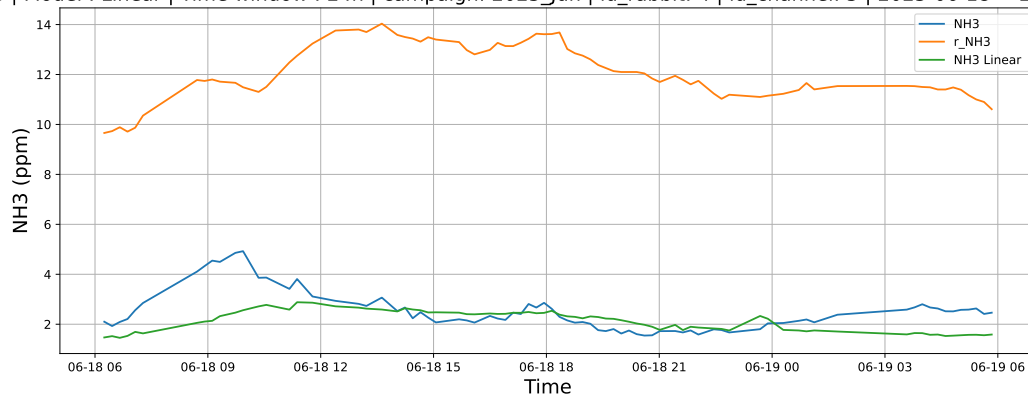
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



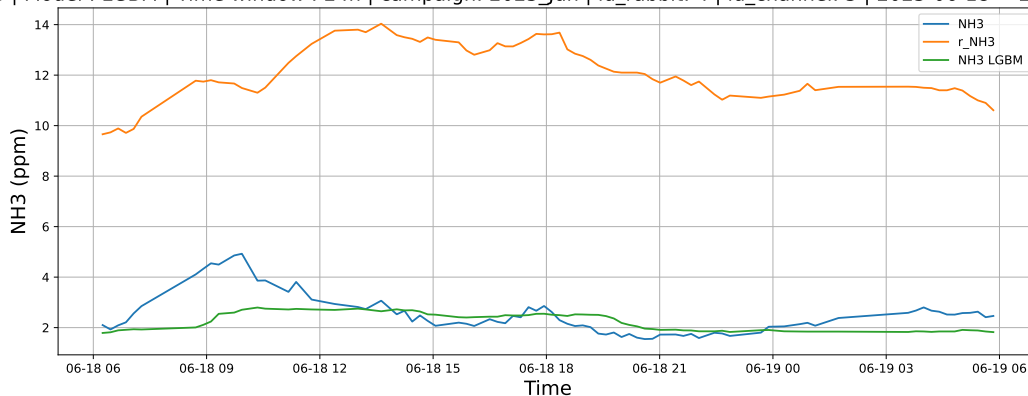
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

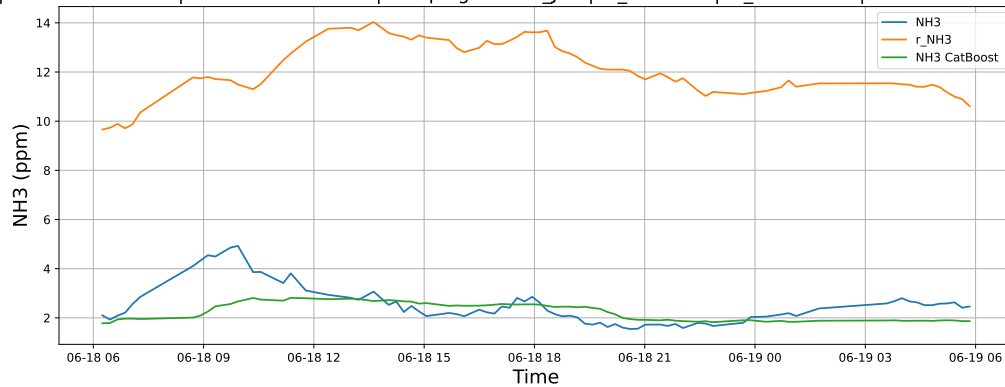
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

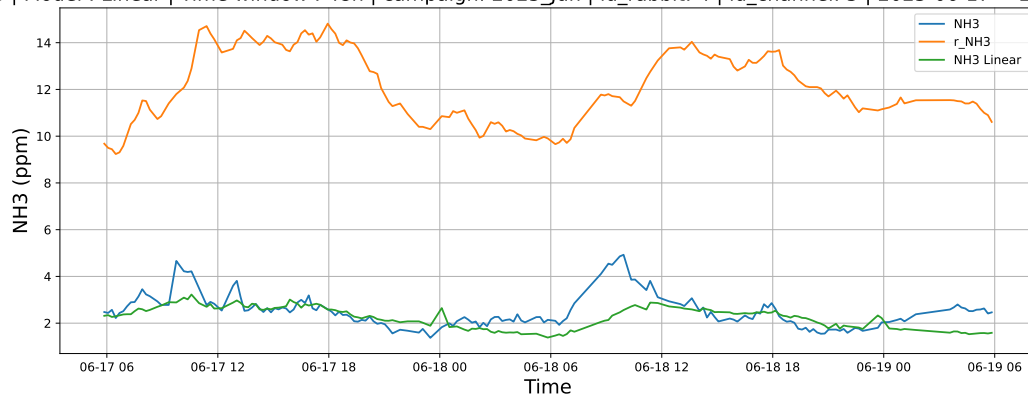


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

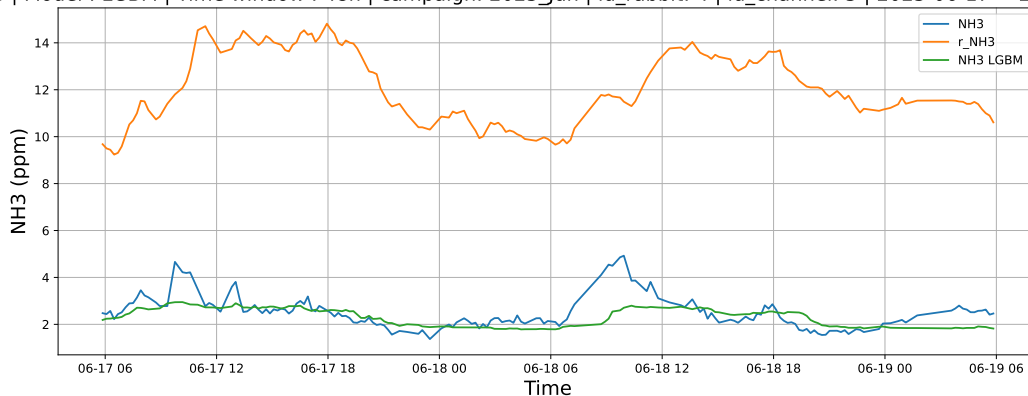


4.1.3.2 Time window : 48

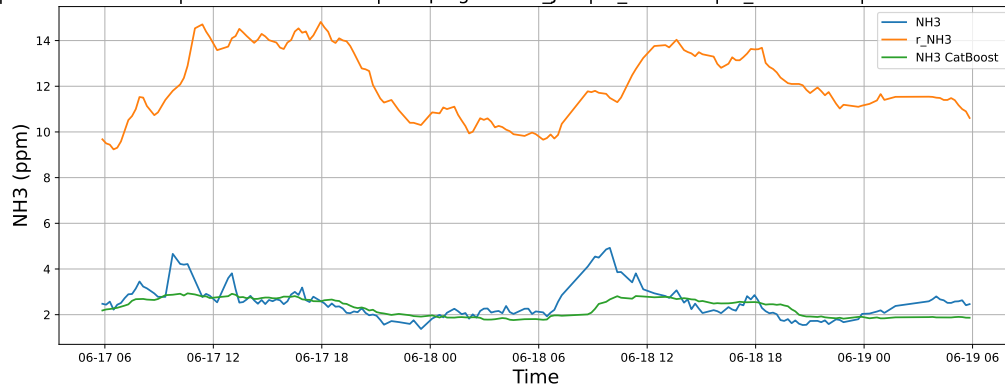
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

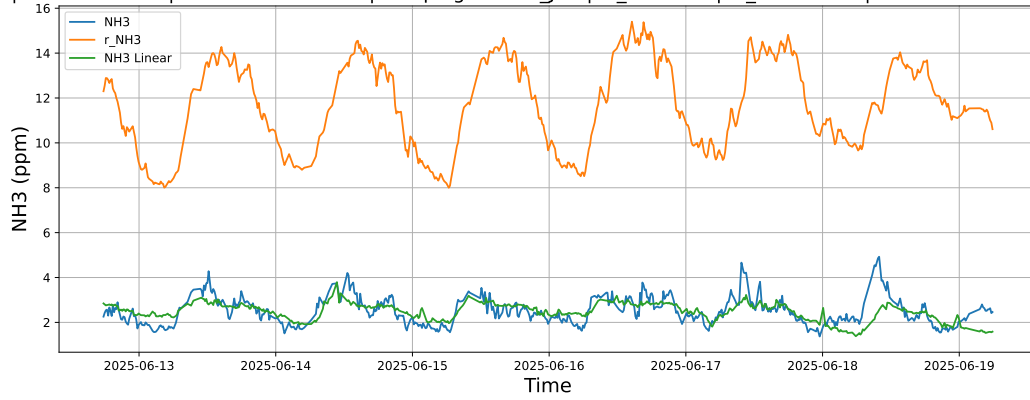


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

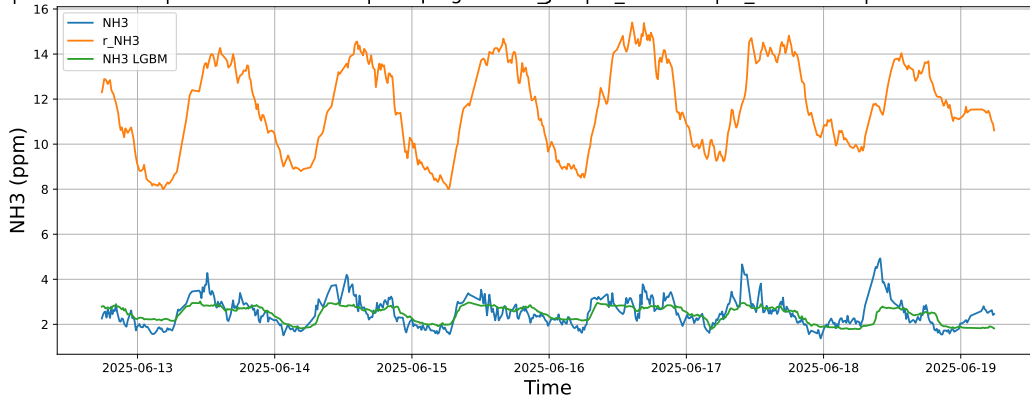


4.1.3.3 Time window : ALL

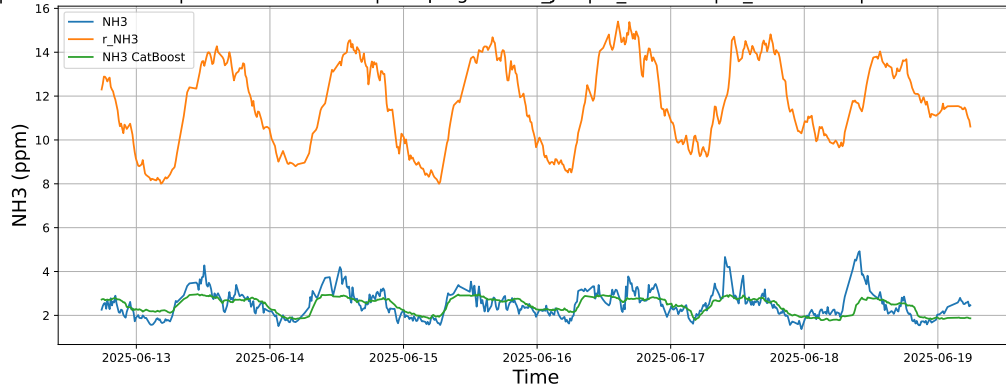
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



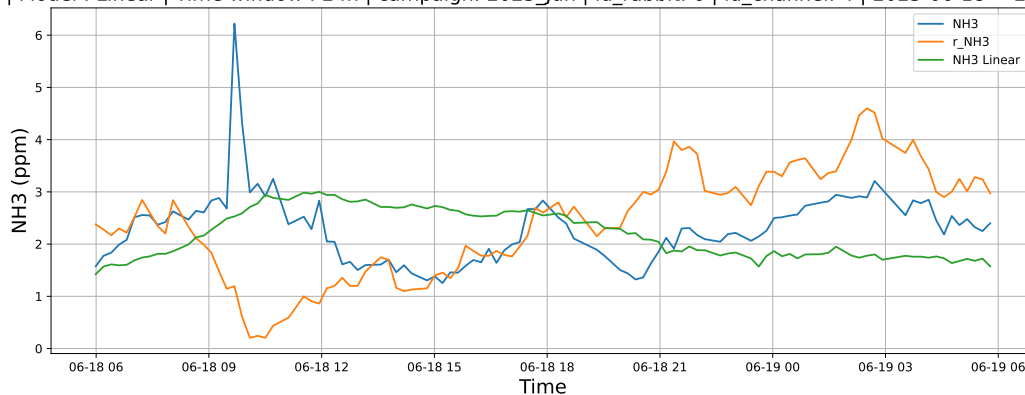
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



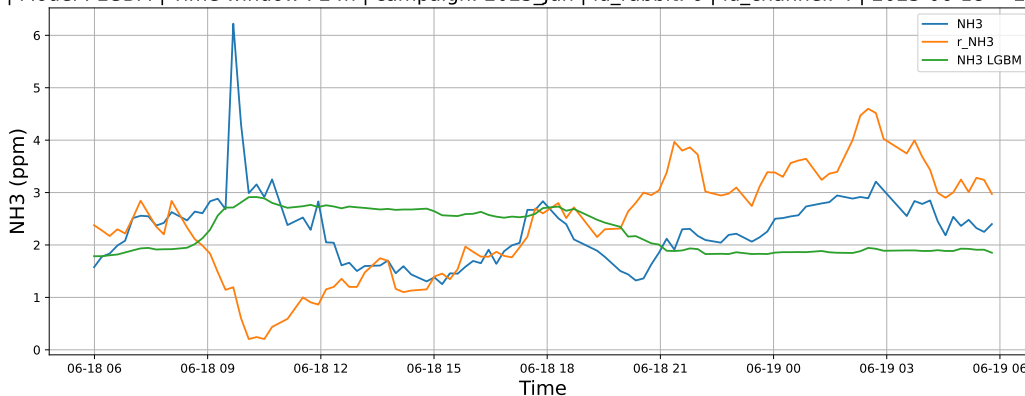
4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

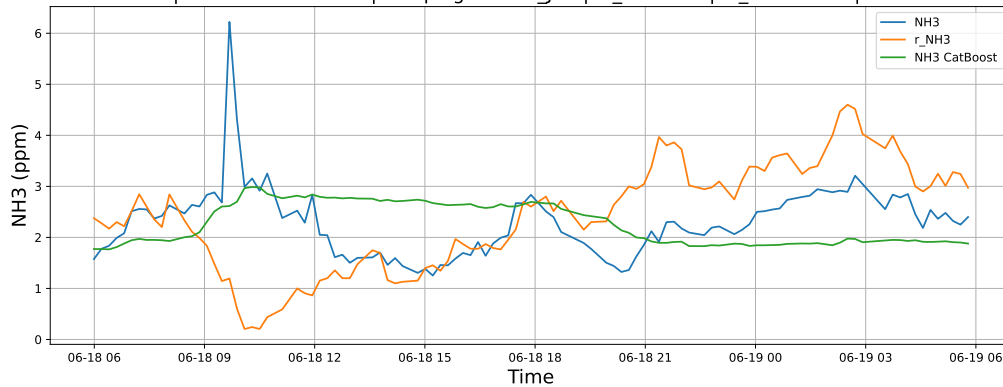
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

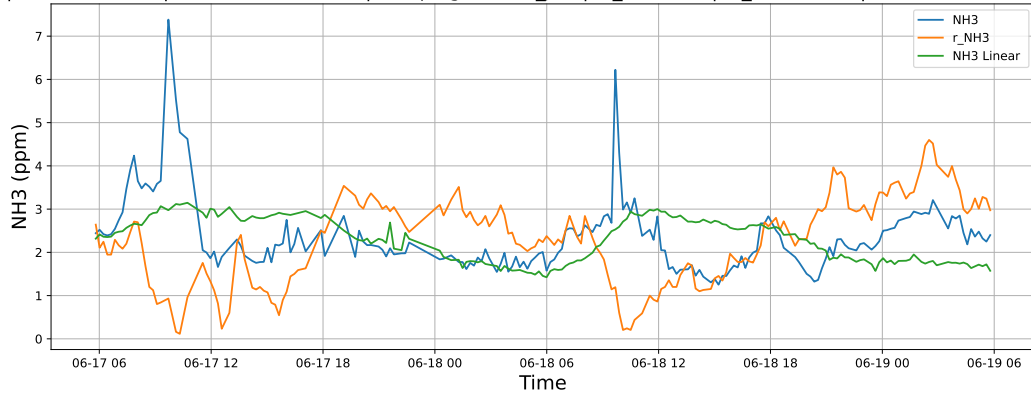


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

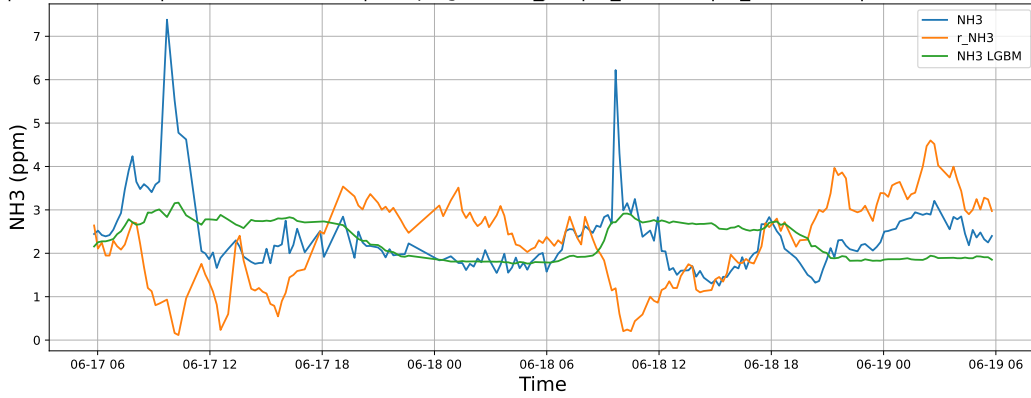


4.1.4.2 Time window : 48

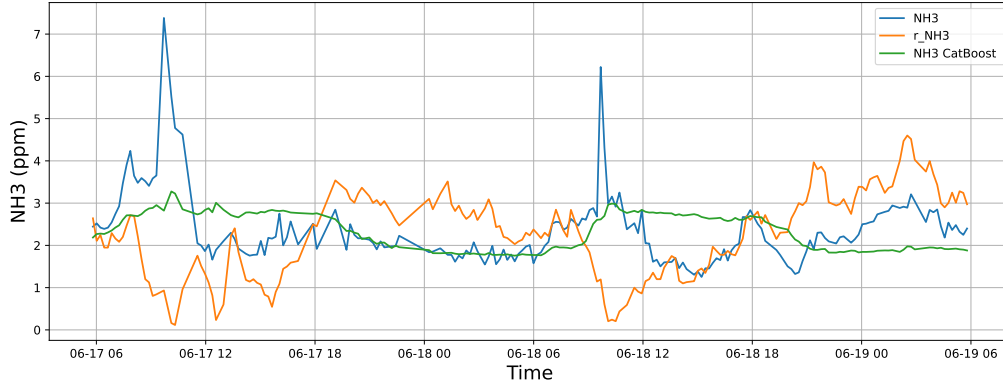
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

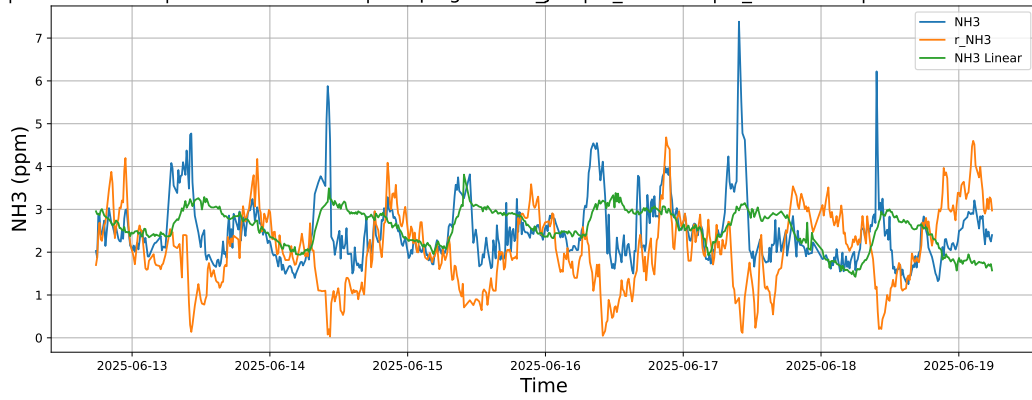


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

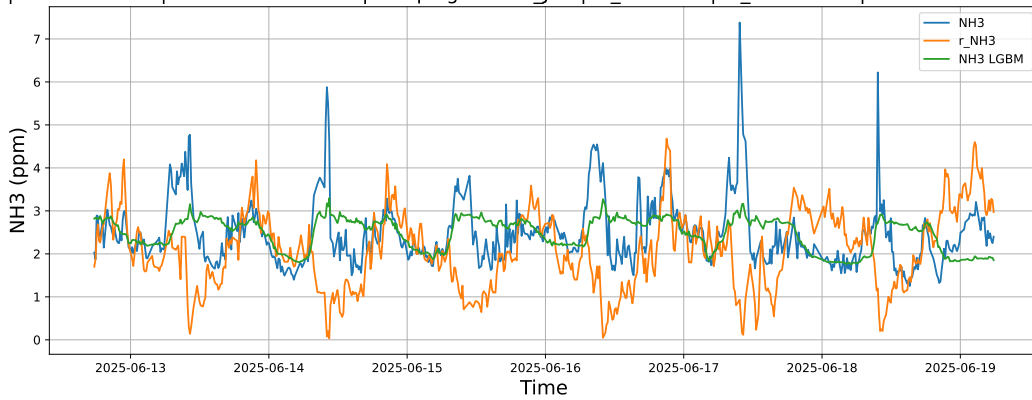


4.1.4.3 Time window : ALL

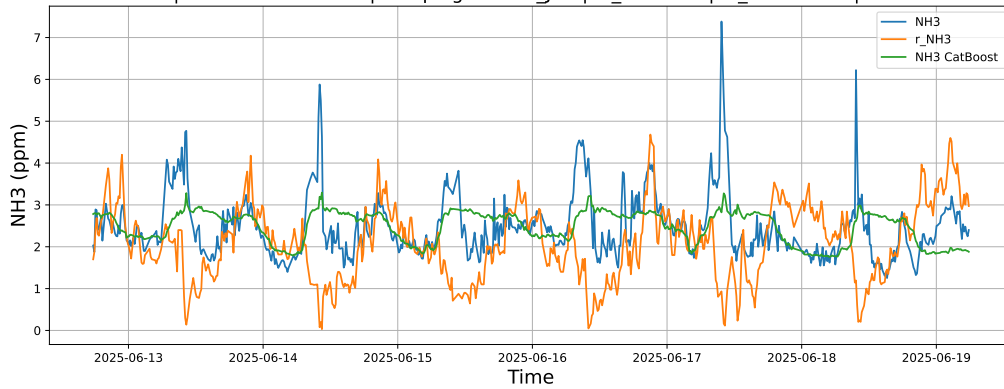
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



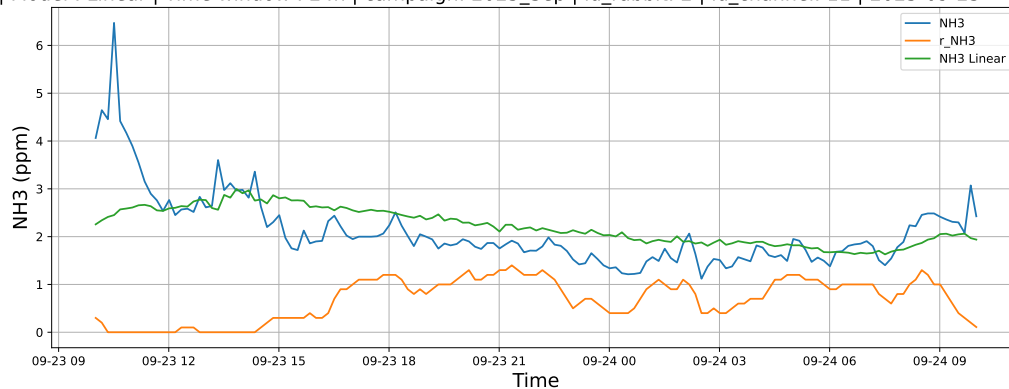
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



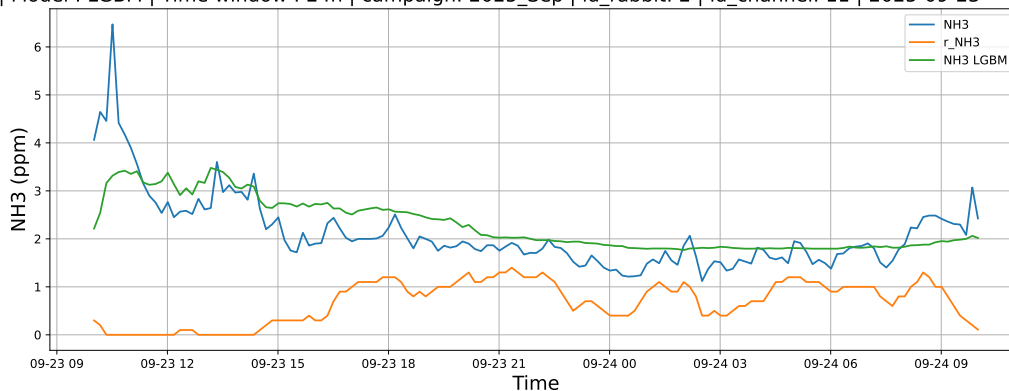
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

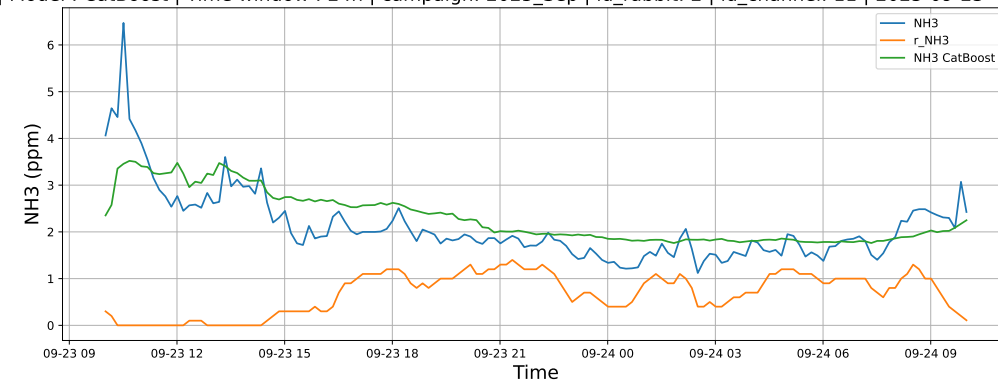
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

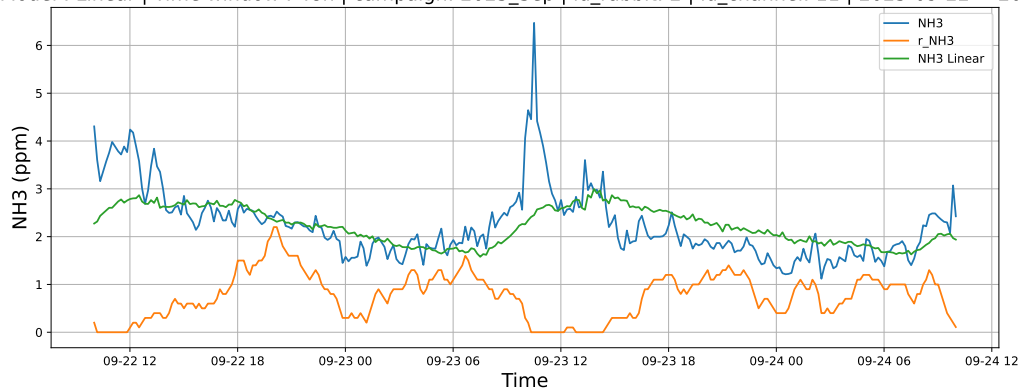


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

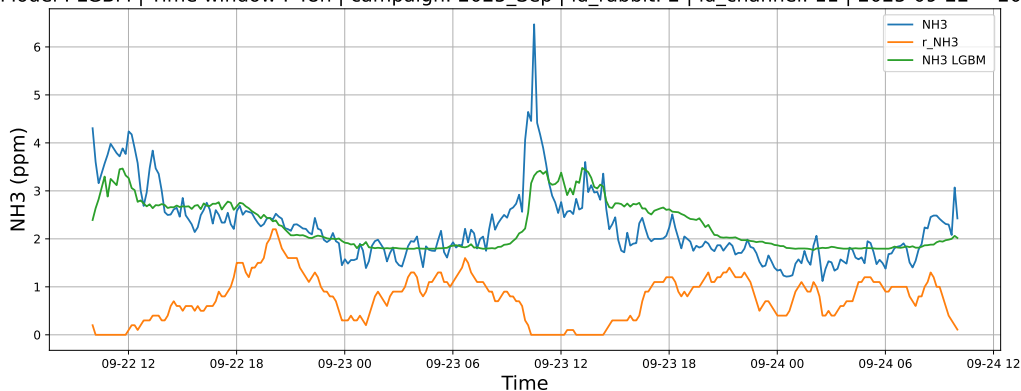


4.1.5.2 Time window : 48

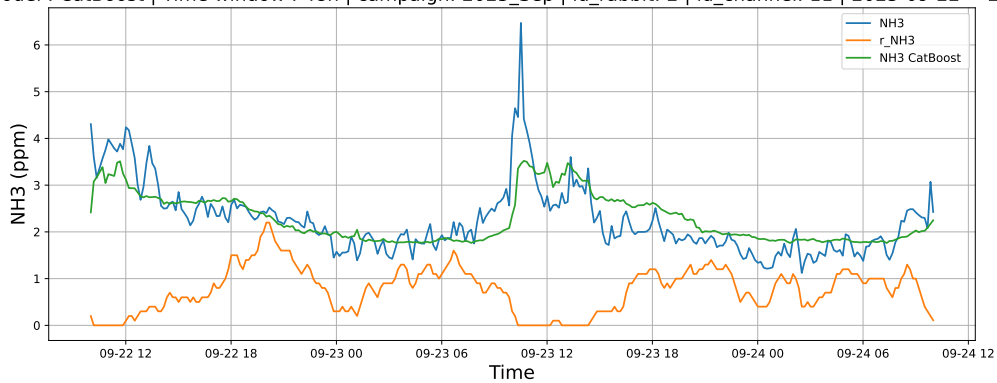
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

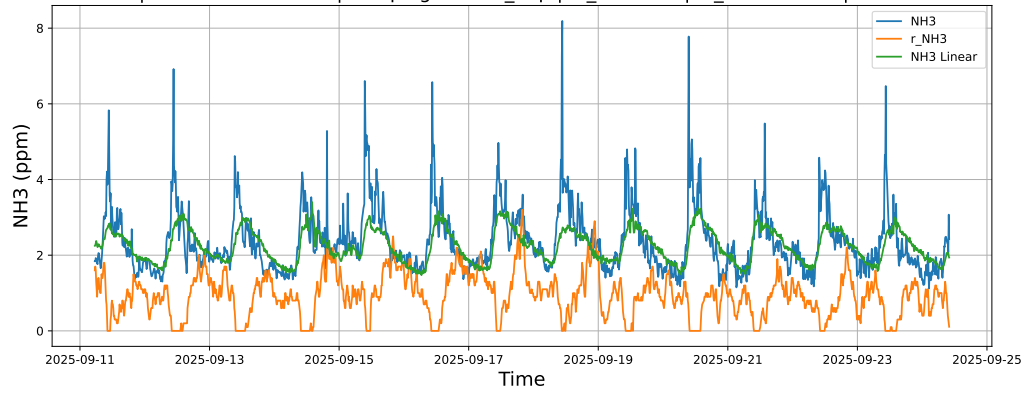


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

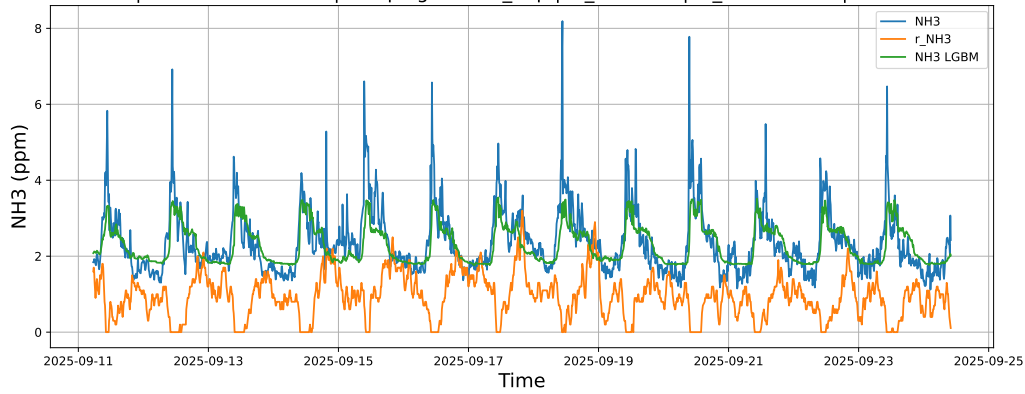


4.1.5.3 Time window : ALL

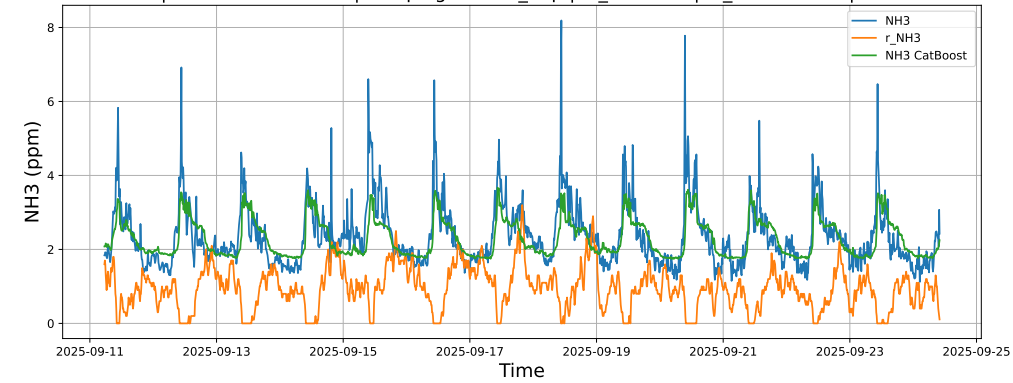
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



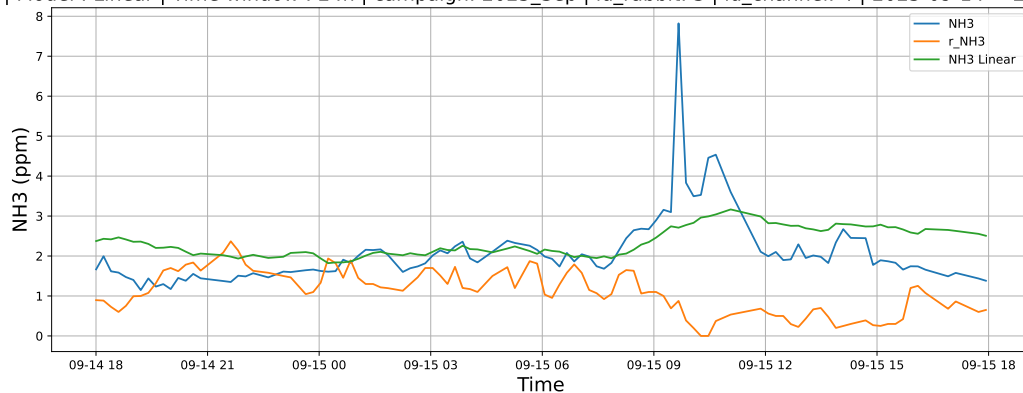
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



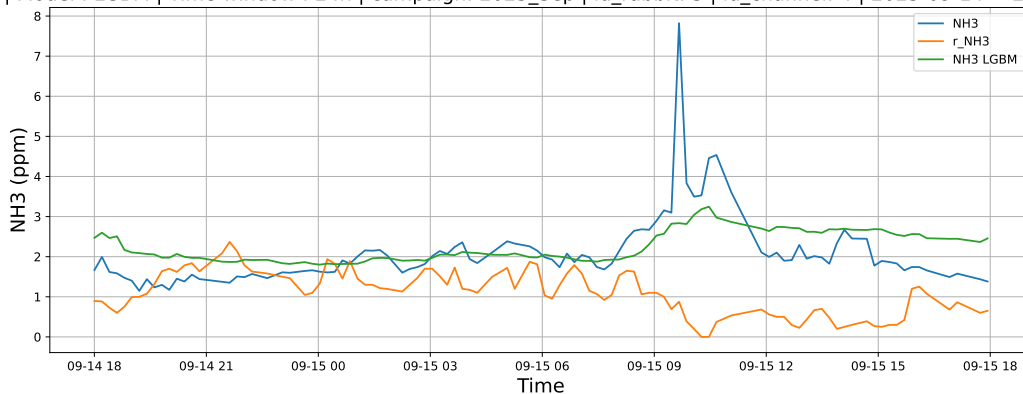
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

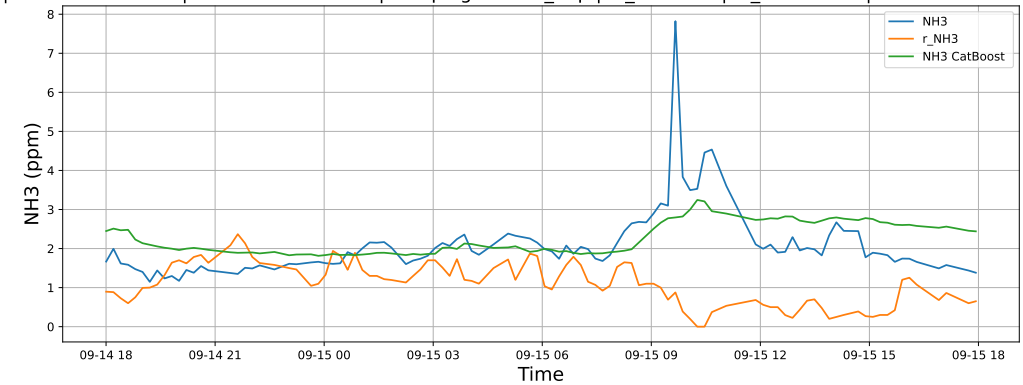
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

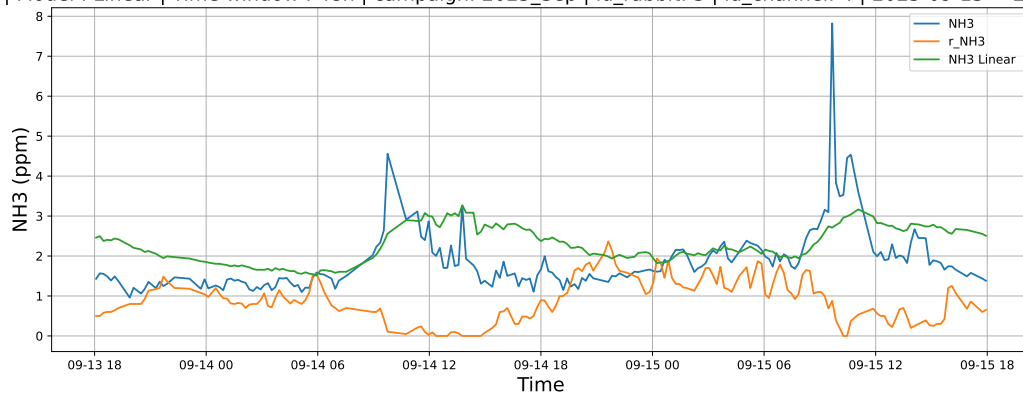


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

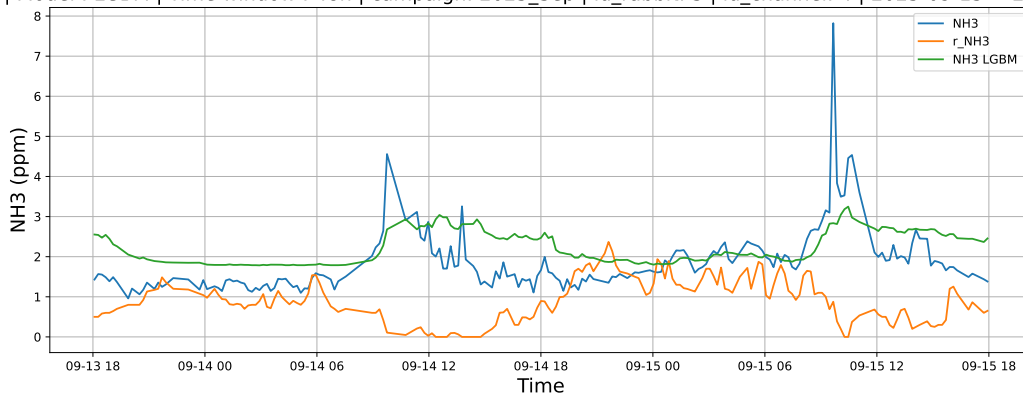


4.1.6.2 Time window : 48

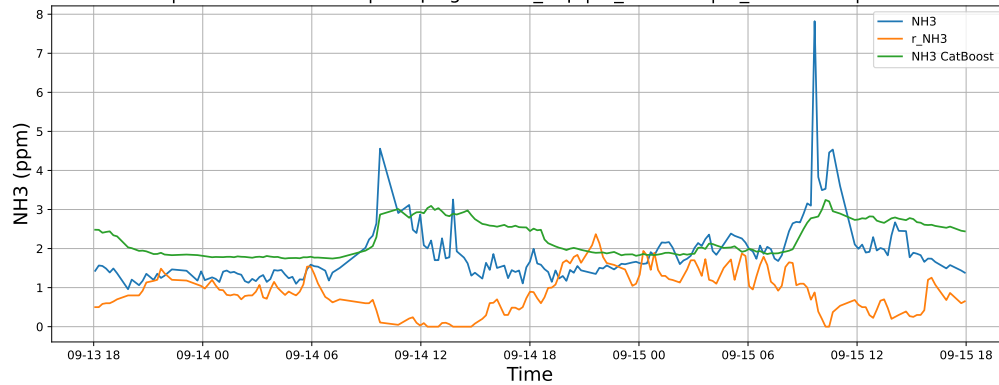
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

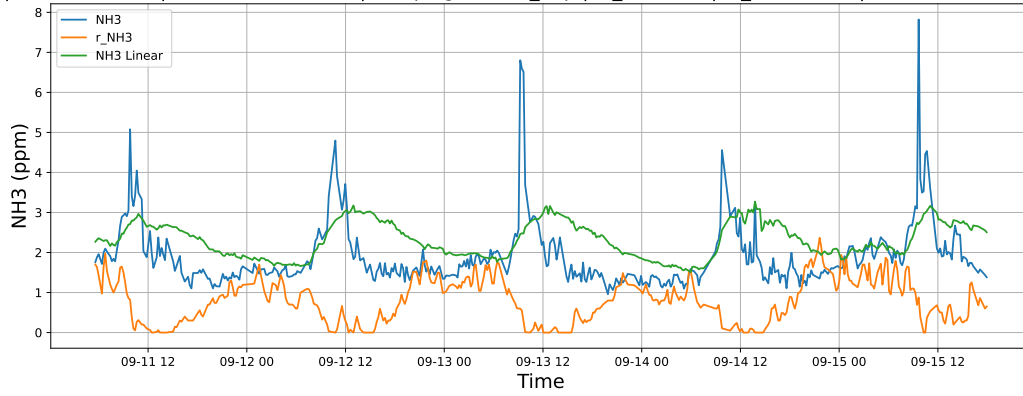


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

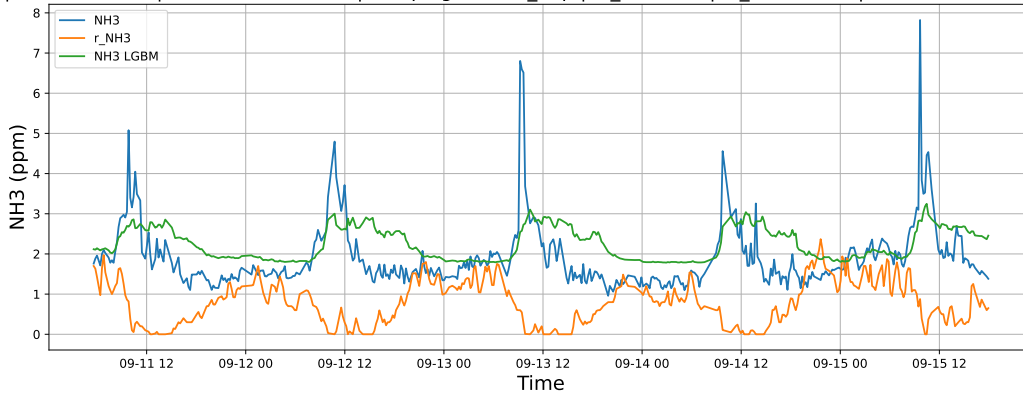


4.1.6.3 Time window : ALL

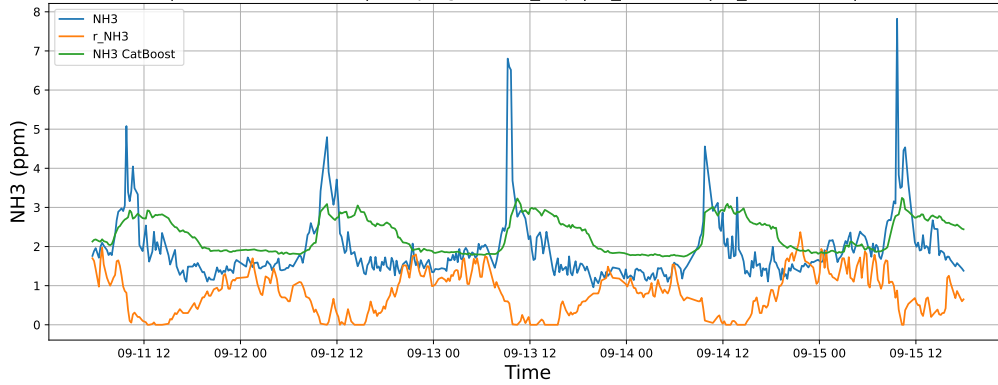
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



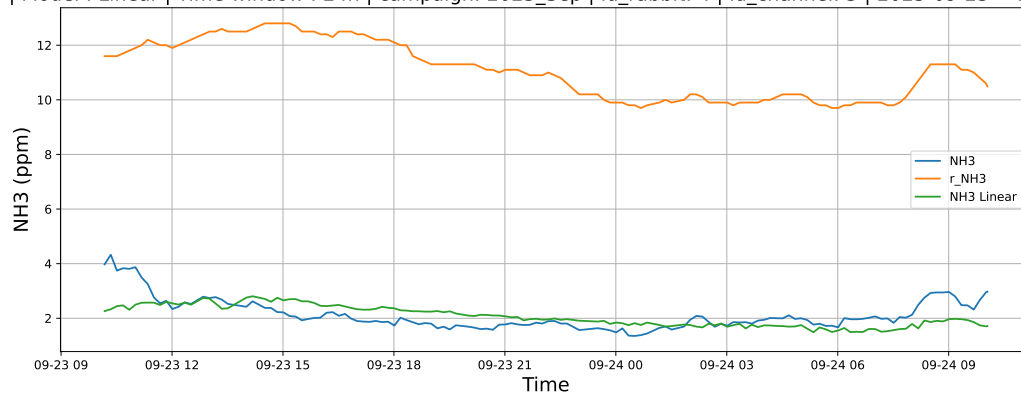
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



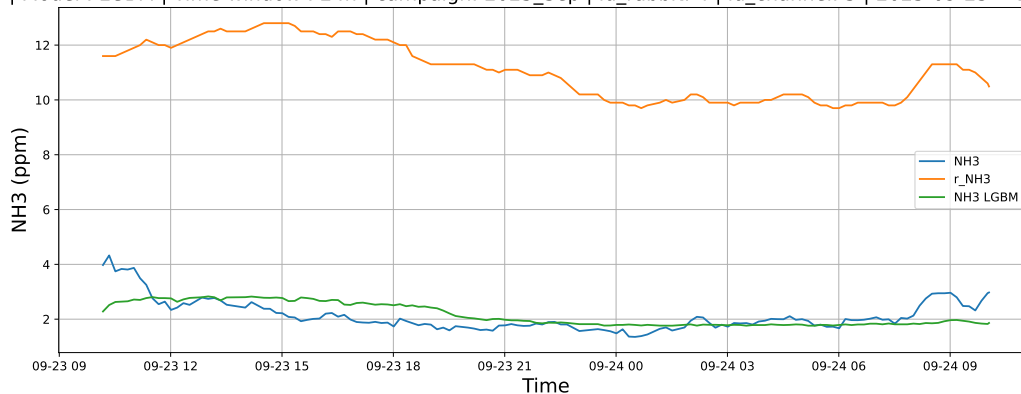
4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

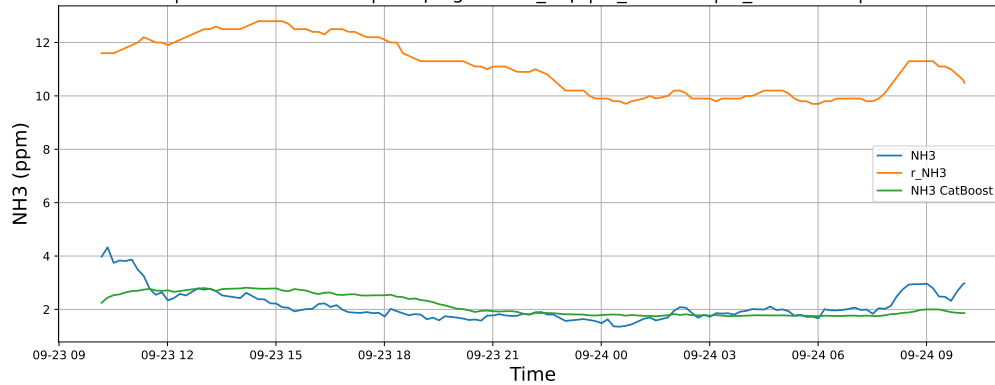
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

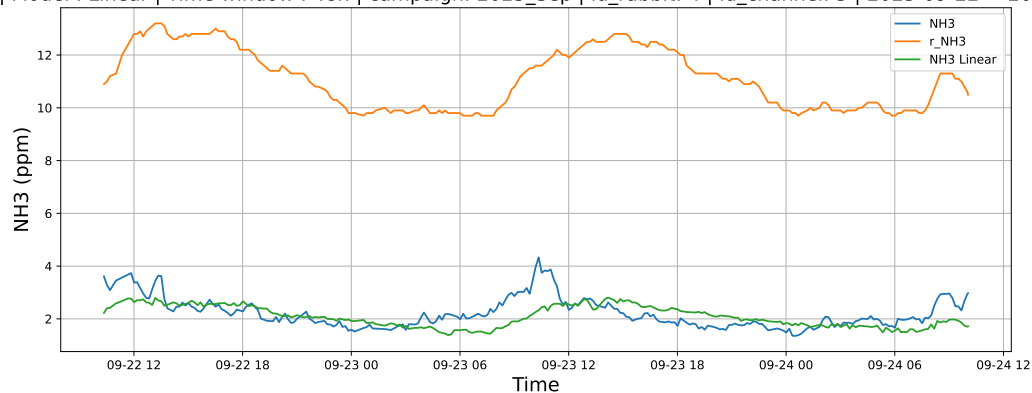


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

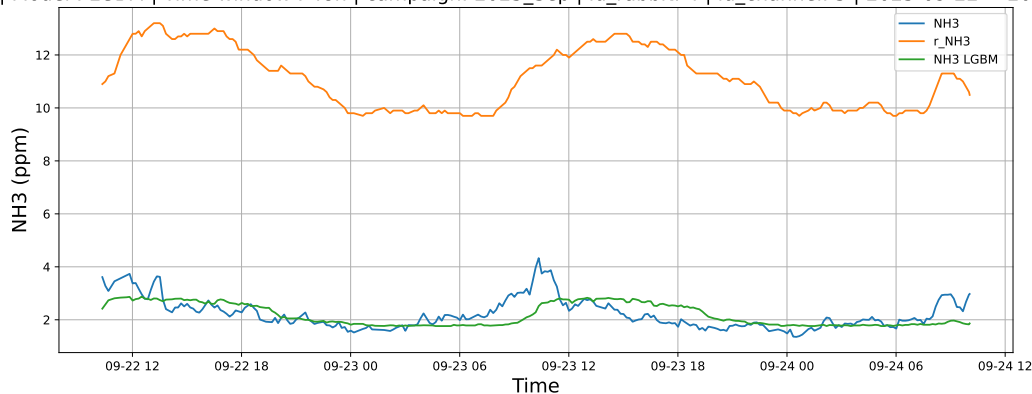


4.1.7.2 Time window : 48

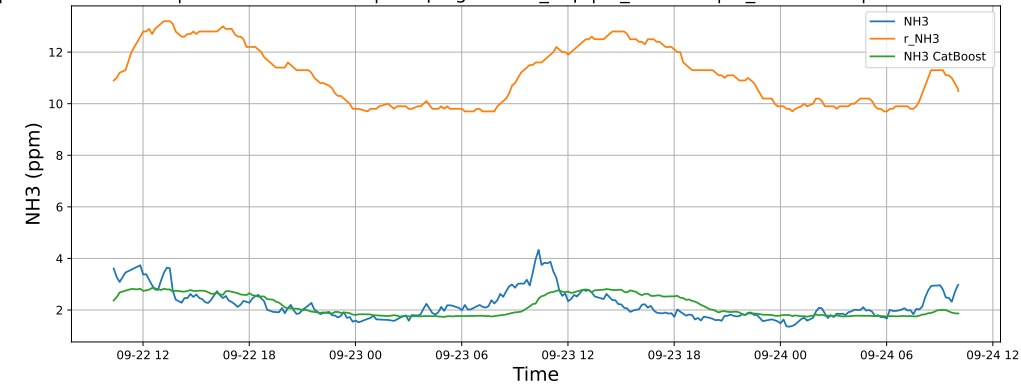
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

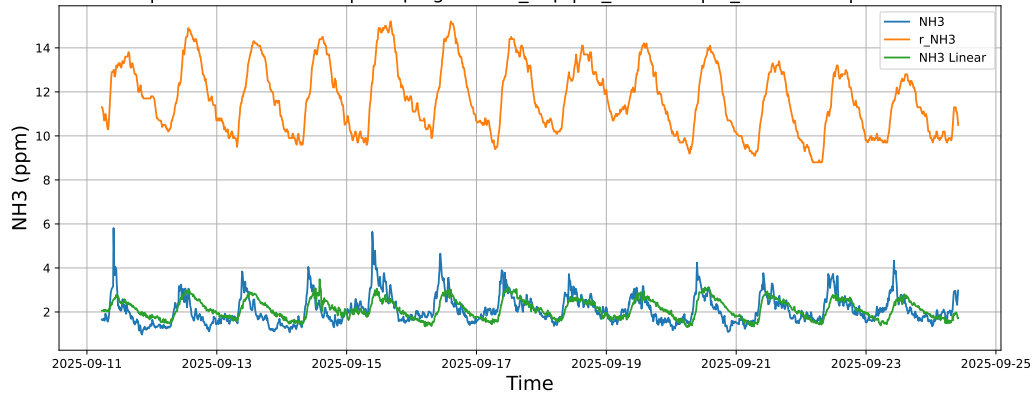


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

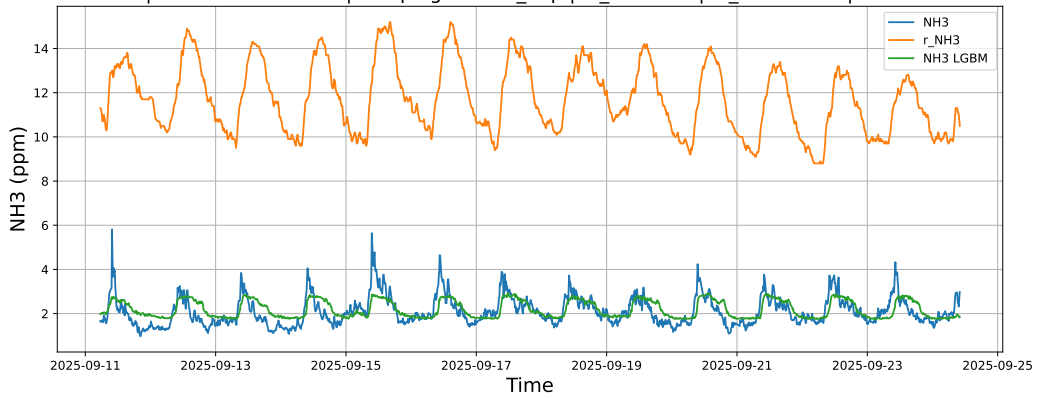


4.1.7.3 Time window : ALL

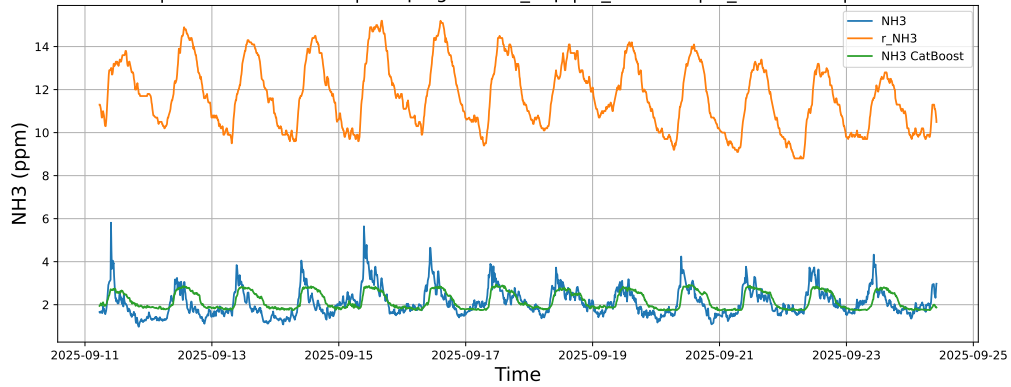
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



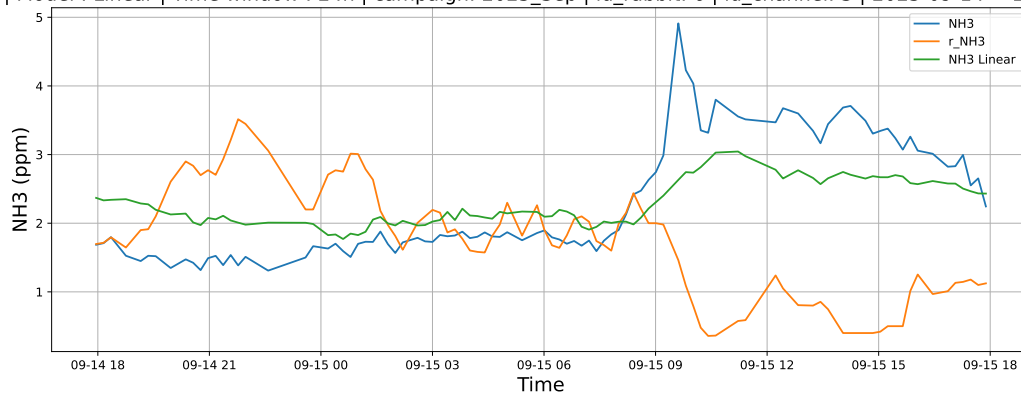
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



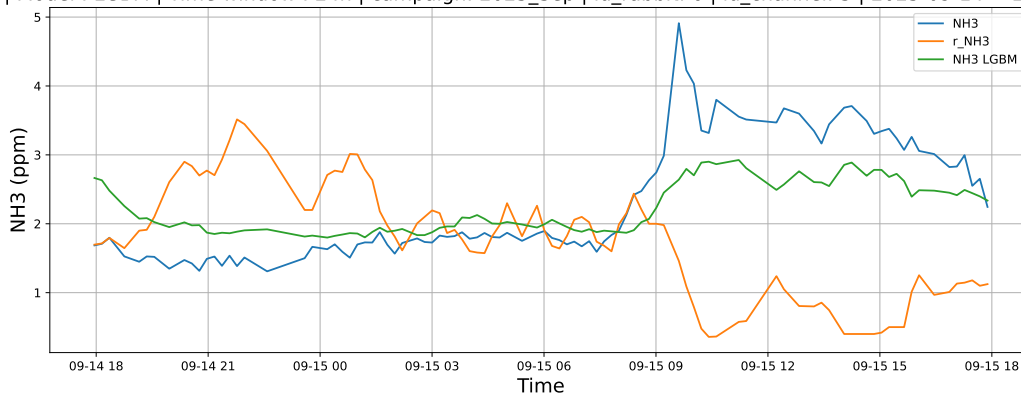
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

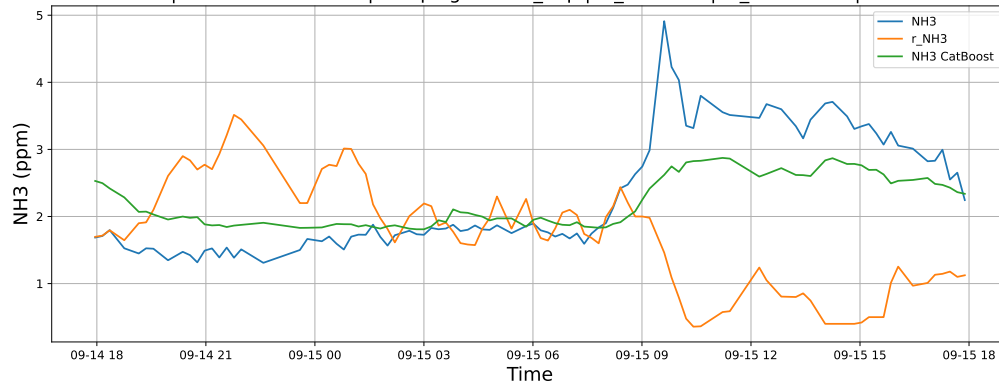
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

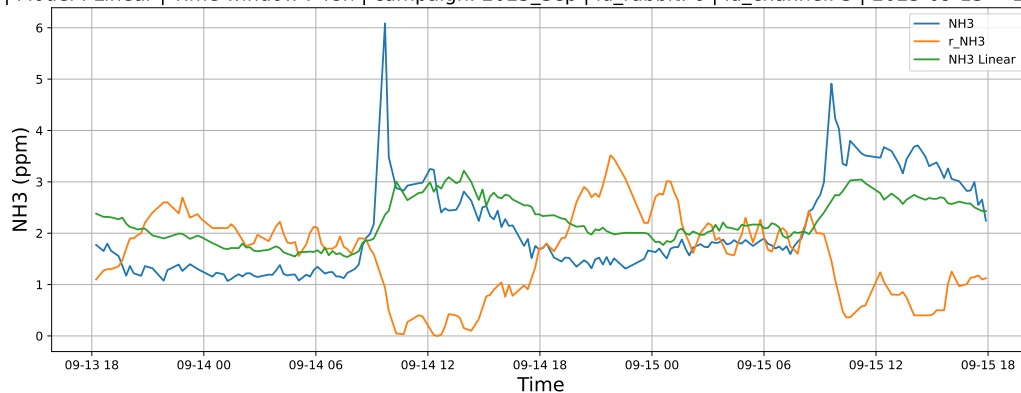


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

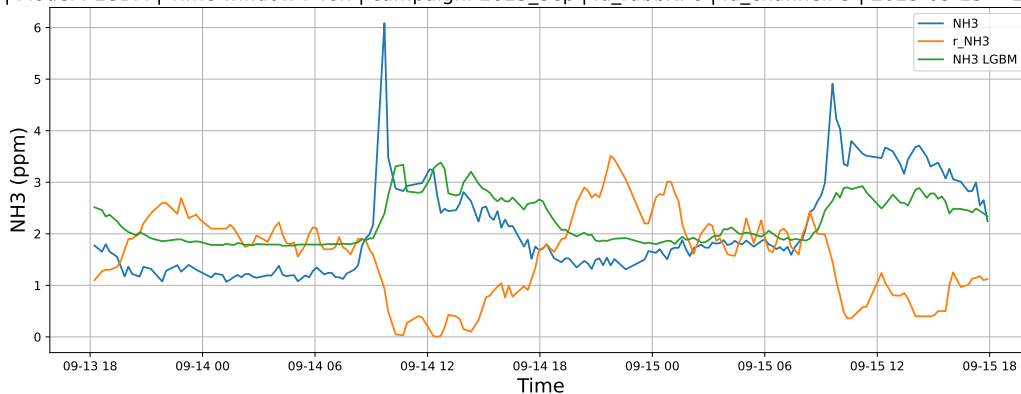


4.1.8.2 Time window : 48

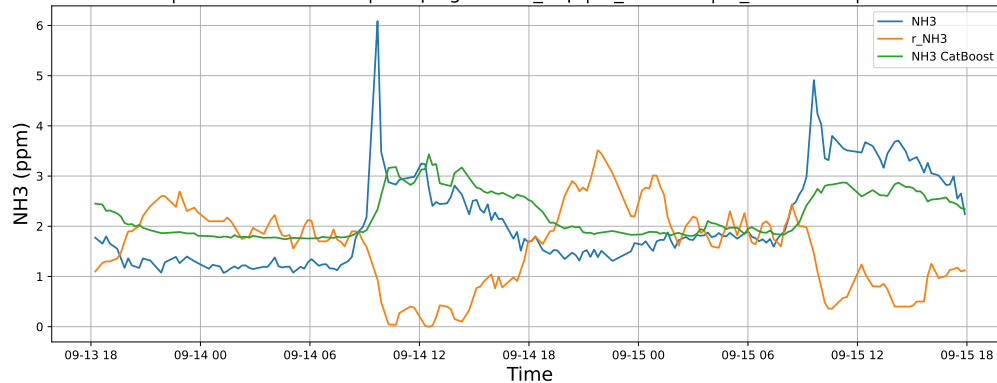
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

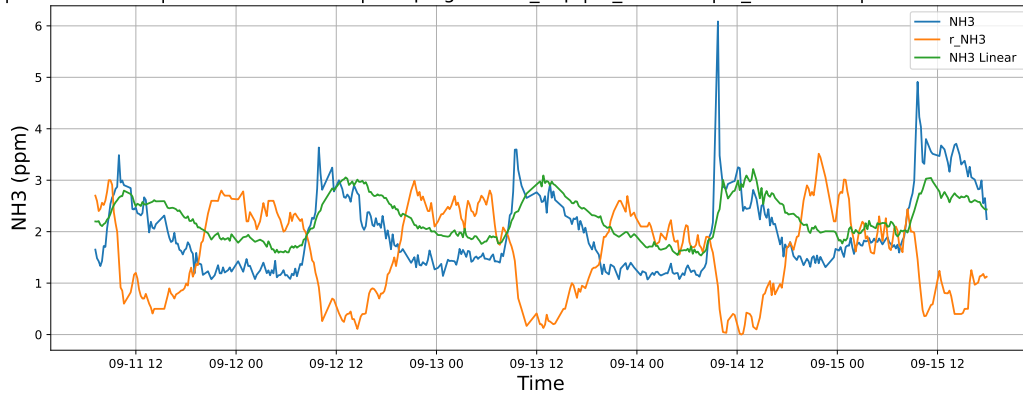


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

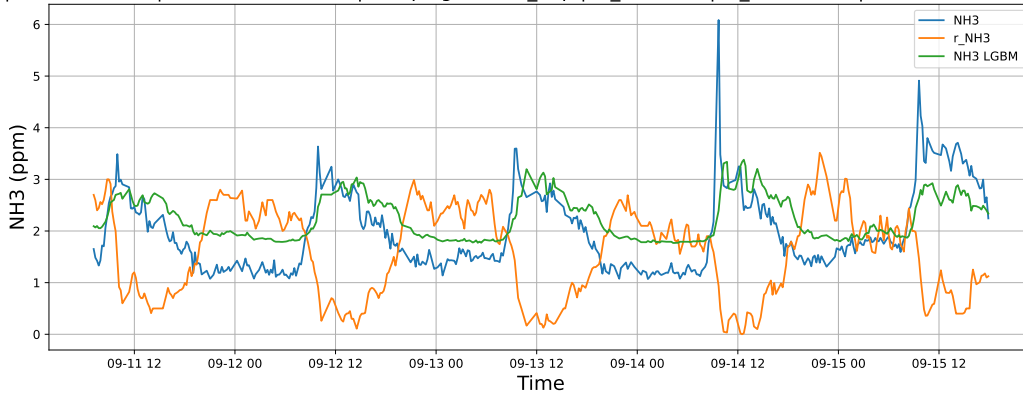


4.1.8.3 Time window : ALL

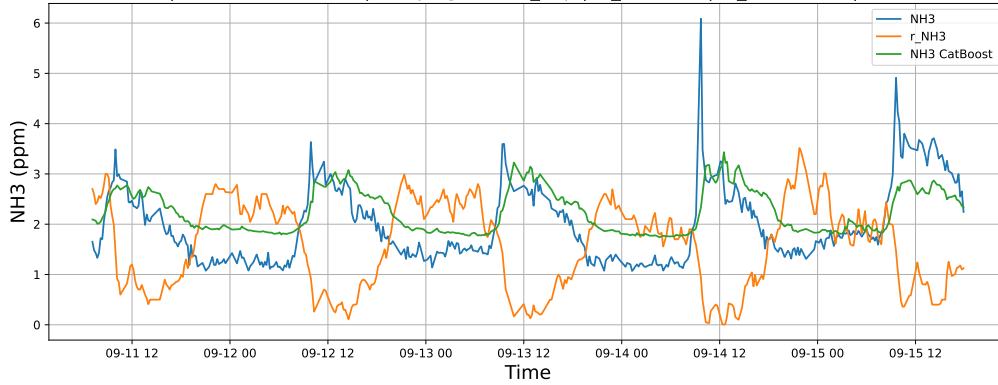
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

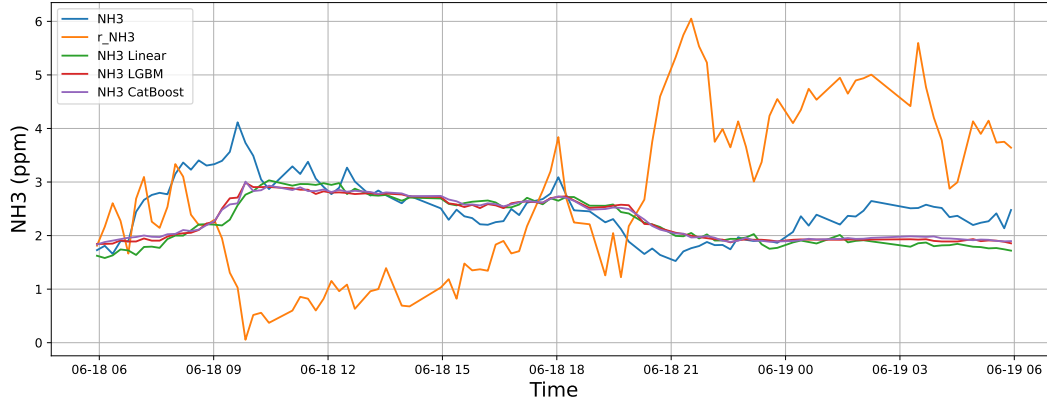


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

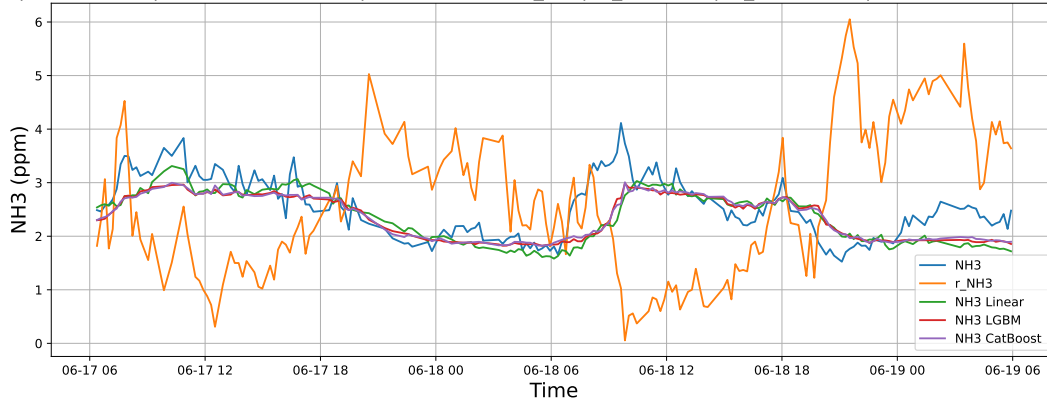
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



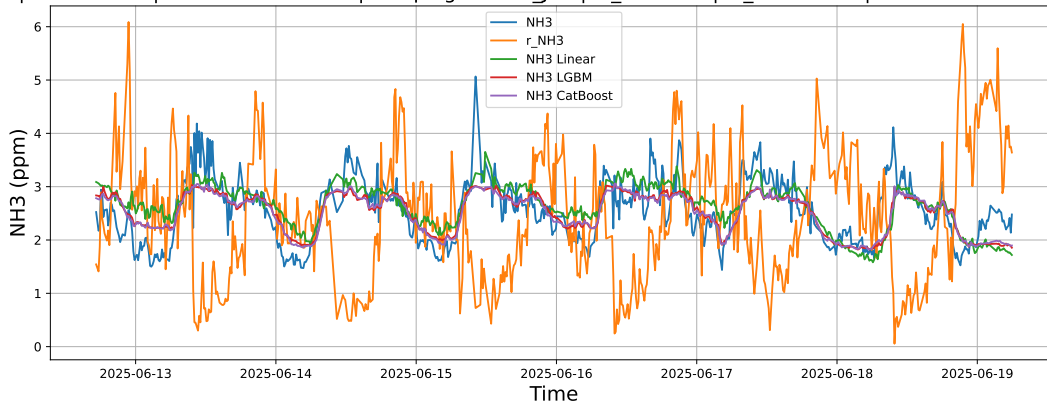
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.2.1.3 Time window : ALL

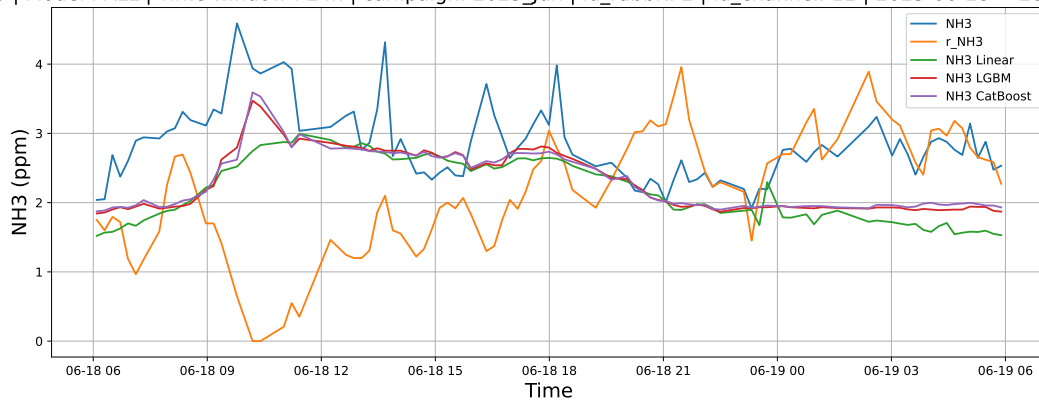
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

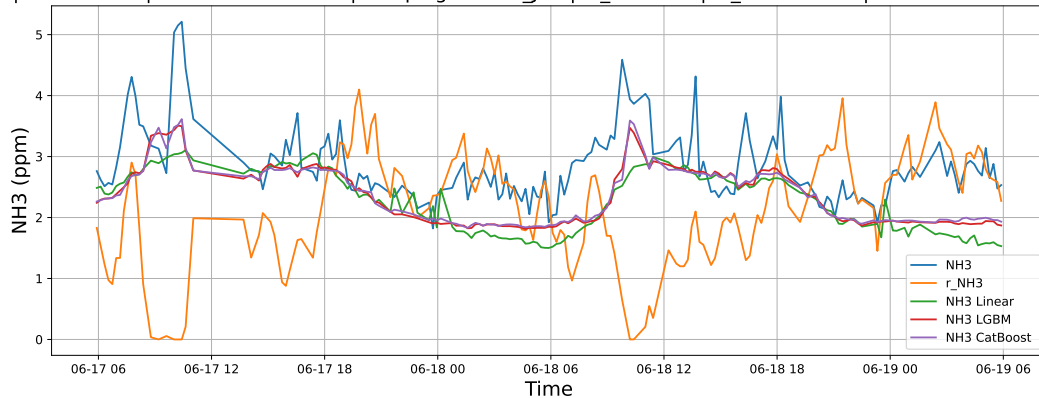
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



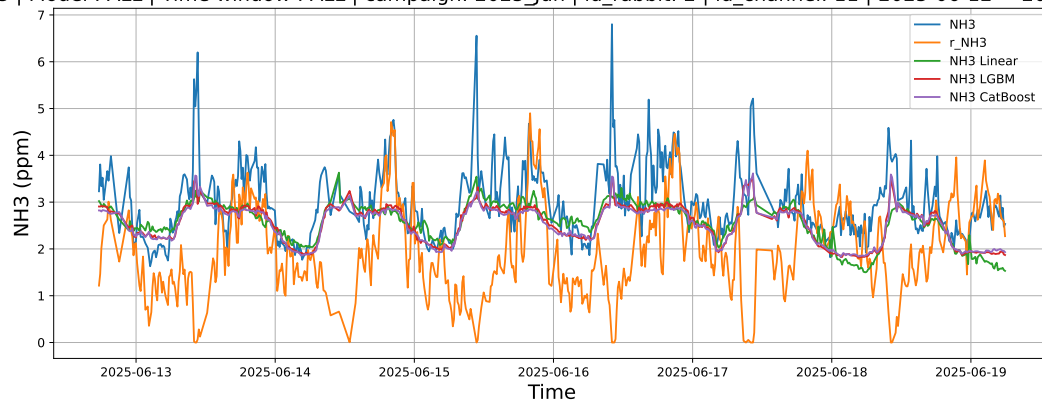
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

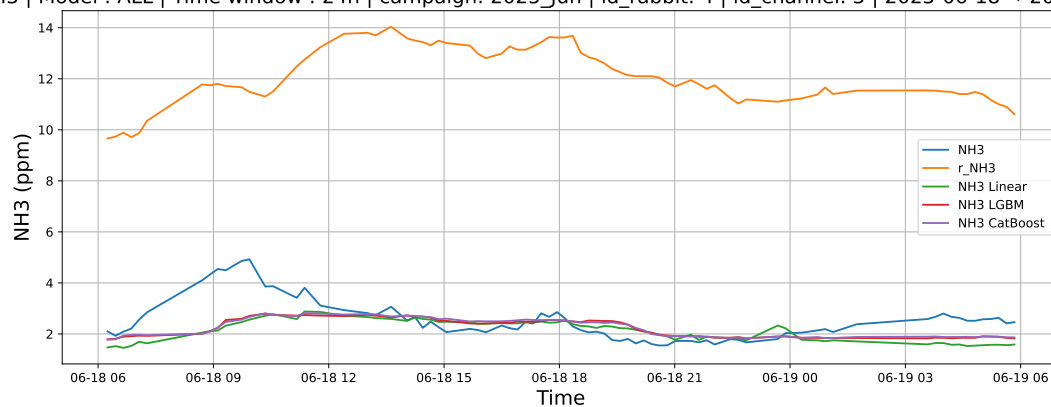
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

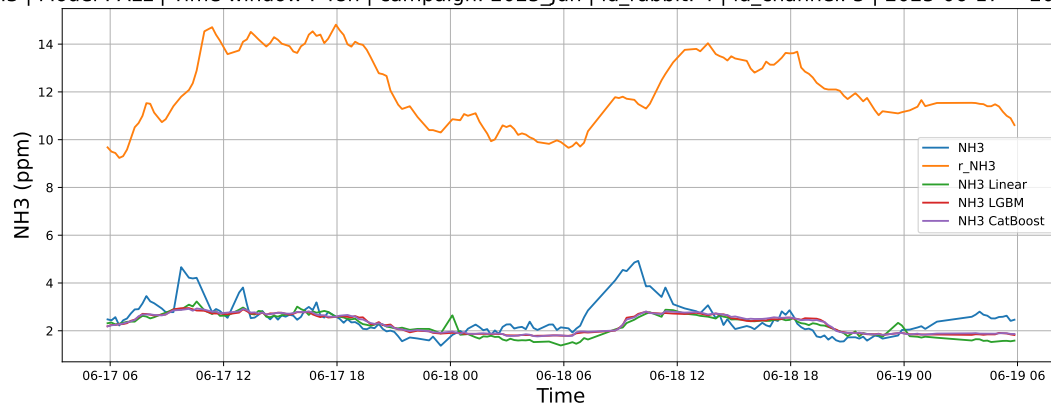
4.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



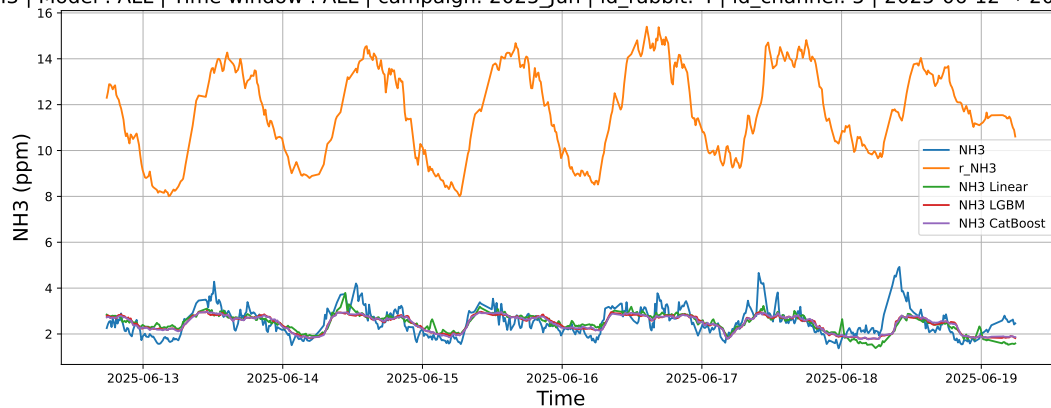
4.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.2.3.3 Time window : ALL

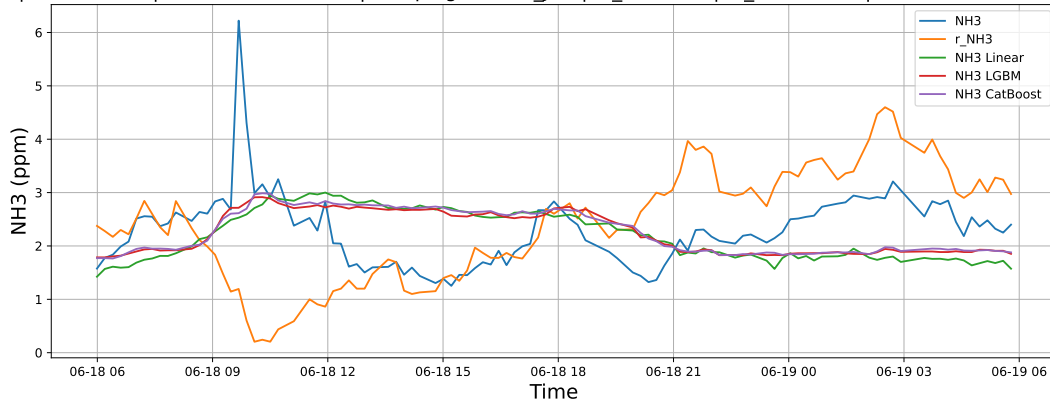
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

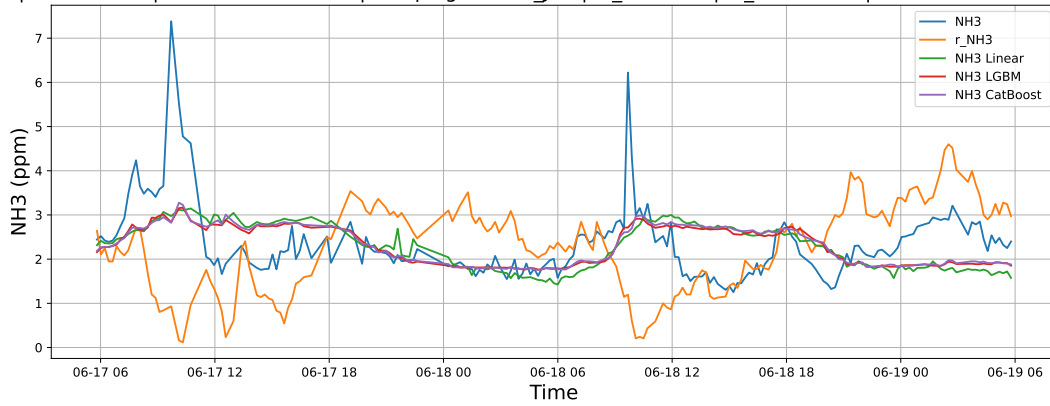
4.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



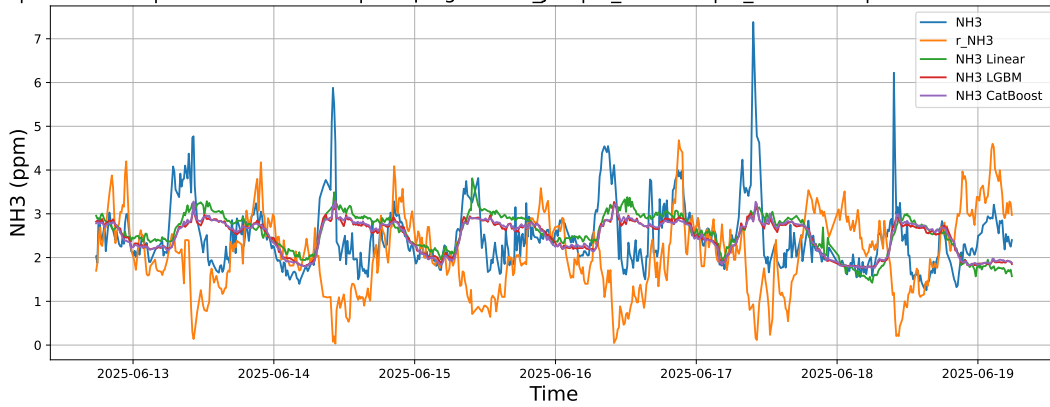
4.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.2.4.3 Time window : ALL

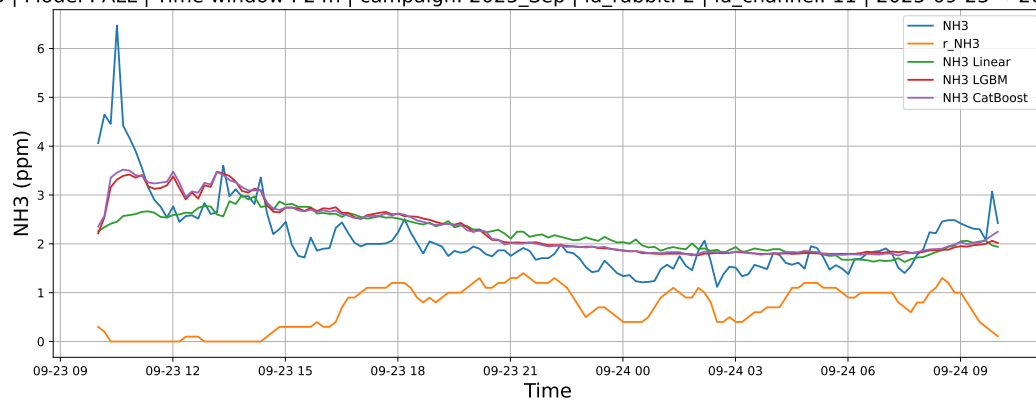
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

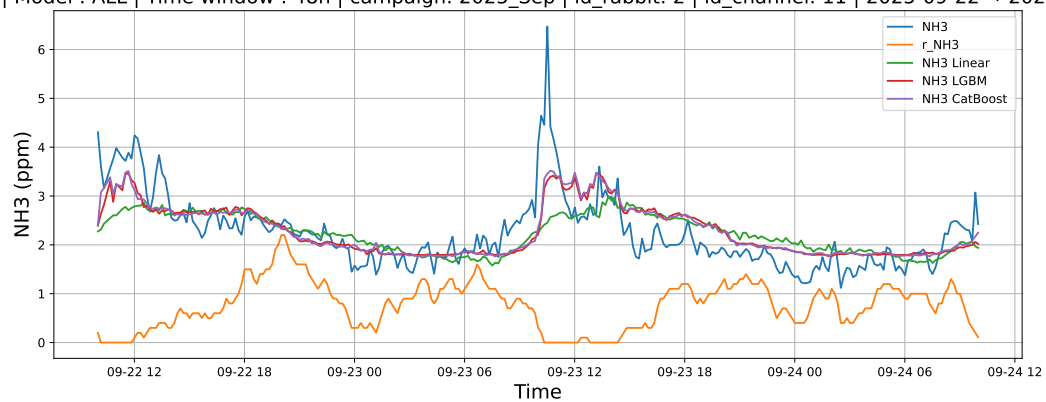
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



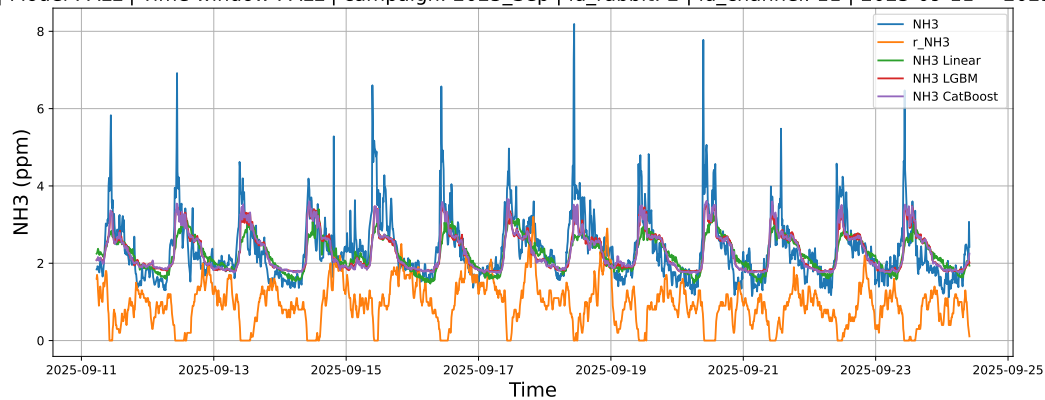
4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.2.5.3 Time window : ALL

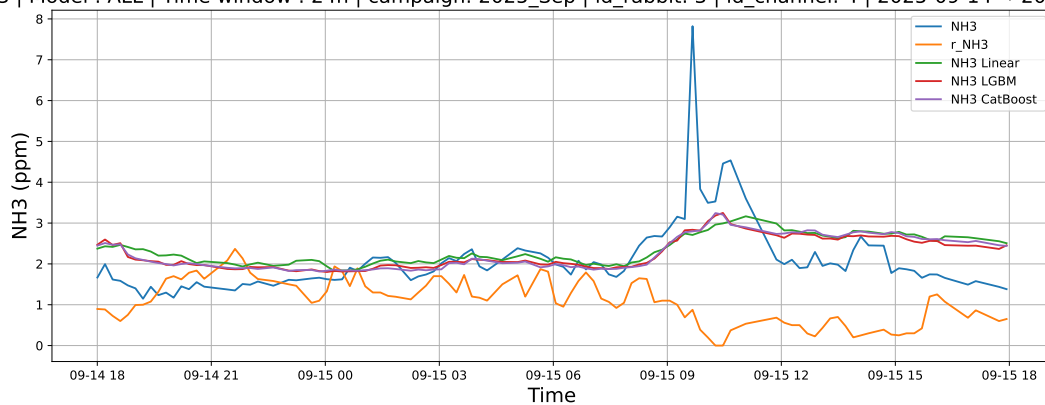
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

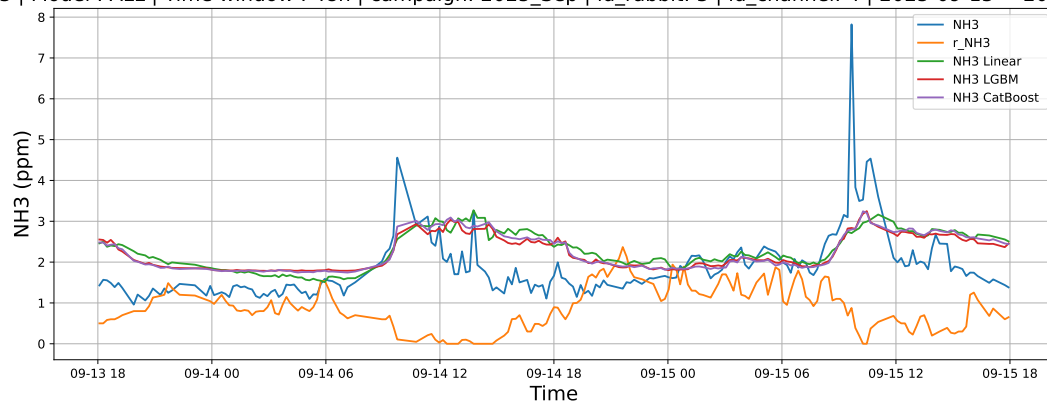
4.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



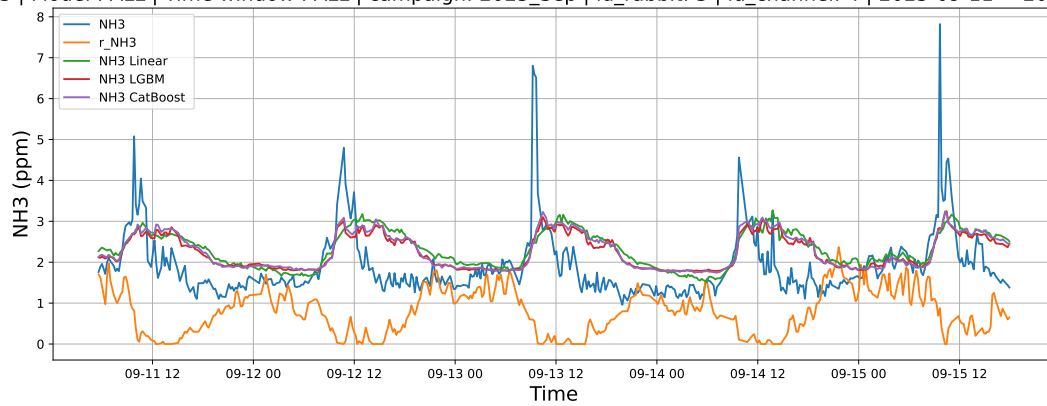
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.2.6.3 Time window : ALL

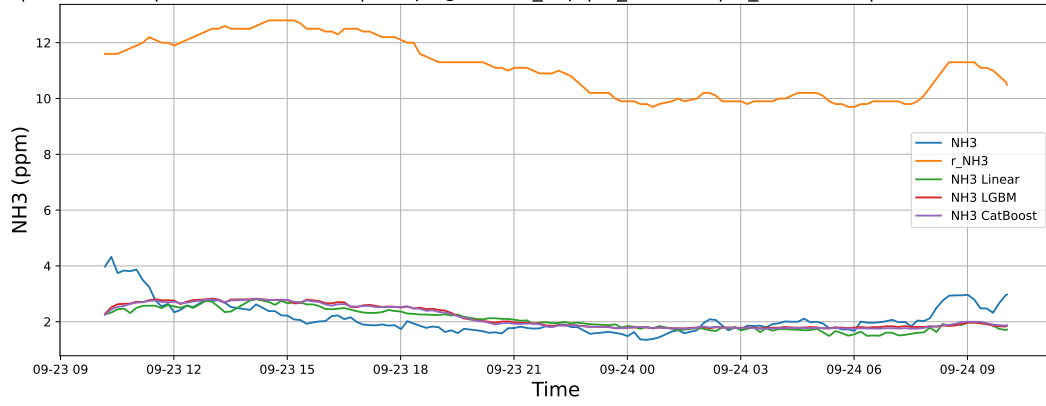
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

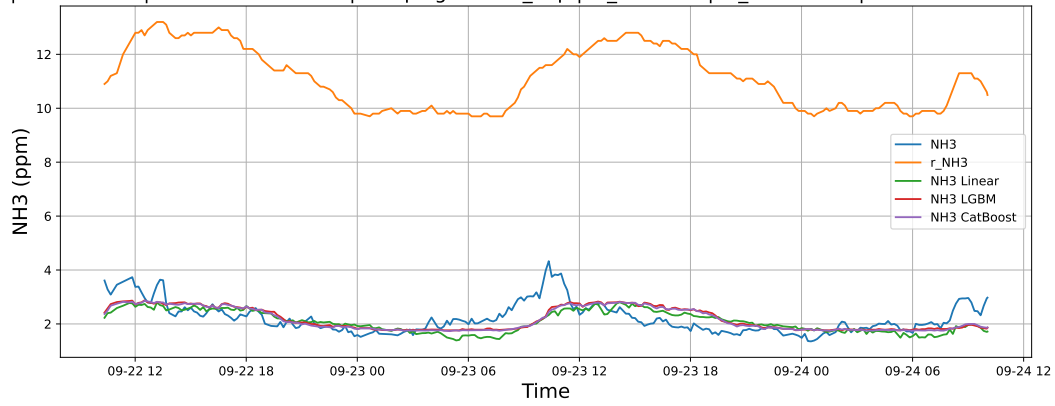
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



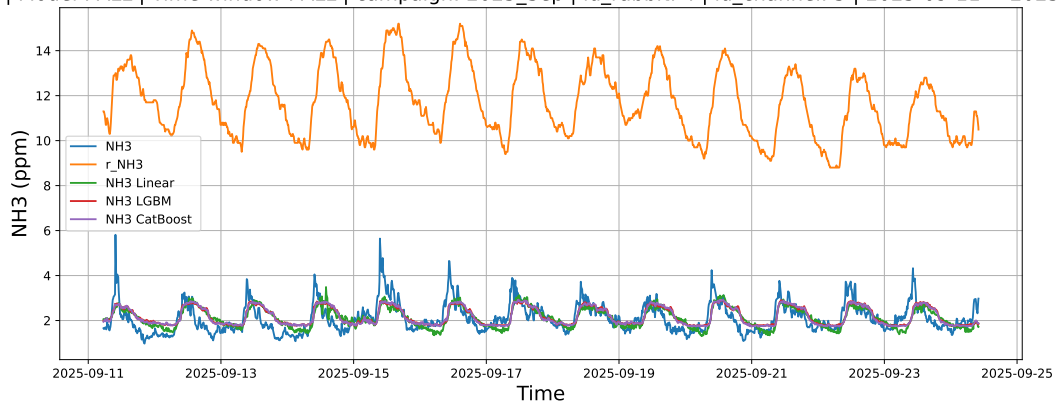
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



4.2.7.3 Time window : ALL

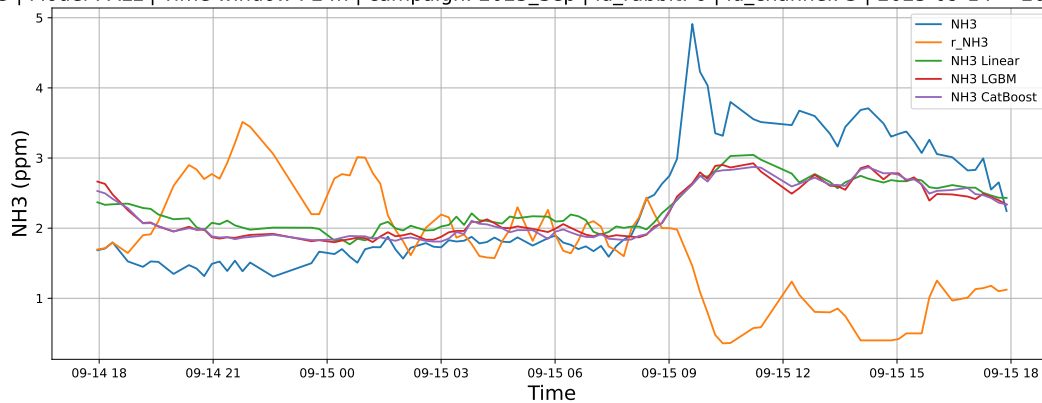
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

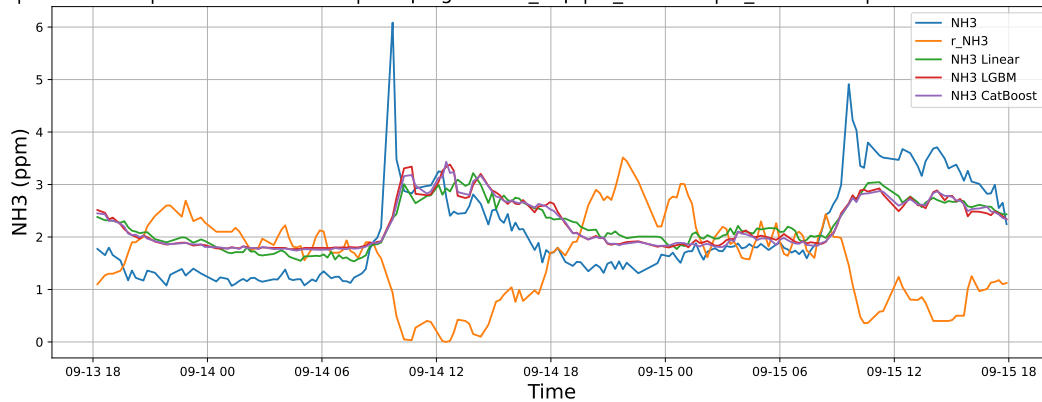
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

